



NetFlow and sFlow Configuration Guide on Cisco 8000 Series Routers, Cisco IOS XR Releases

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CHAPTER 1

About NetFlow and sFlow Documentation

Introducing the updated *NetFlow and sFlow Configuration on Cisco 8000 Series Routers*, Cisco IOS XR Releases.

Release-agnostic Document

This document for the Cisco 8000 Series Router features a single version that will be consistently kept up to date with the latest features and releases. Our goal is to make it easier for you to bookmark a single link and find one comprehensive version of the Netflow and sFlow Configuration for the 8000 Series Routers, rather than sift through multiple versions that are specific to each IOS XR release.

Where to begin

To get started with the NetFlow and sFlow implementation, we recommend referring to the [Cisco IOS XR Feature, Release, and Platform Matrix](#). This section offers insights into NetFlow and sFlow features introduced not only for the 8000 Series Routers but also for all other IOS XR Routing platforms. Our aim is to simplify your access to our documentation and help you develop a comprehensive understanding.

- [Releases Supported, on page 1](#)

Releases Supported

The document is relevant for the following releases:

- IOS XR Release 24.1.1
- [IOS XR Release 7.11.1](#)
- [IOS XR Release 7.10.1](#)
- [IOS XR Release 7.9.2](#)
- [IOS XR Release 7.9.1](#)
- [IOS XR Release 7.8.2](#)
- [IOS XR Release 7.8.1](#)
- [IOS XR Release 7.7.2](#)
- [IOS XR Release 7.7.1](#)

- [IOS XR Release 7.5.3](#)
- [IOS XR Release 7.5.2](#)
- [IOS XR Release 7.5.1](#)
- [IOS XR Release 7.3.4](#)
- [IOS XR Release 7.3.3](#)
- [IOS XR Release 7.3.2](#)
- [IOS XR Release 7.3.15](#)
- [IOS XR Release 7.3.1](#)

Cisco IOS XR Releases prior to those listed here have reached the End of Extended SW Maintenance Date. To access the notification, please visit the [End-of-Life and End-of-Sale Notices page](#).



CHAPTER 2

Features Introduced in Cisco IOS XR Products and Releases

This table summarizes the features enhanced and introduced for NetFlow and sFlow.

- [Feature, Release, and Platform Matrix for NetFlow and sFlow](#), on page 3

Feature, Release, and Platform Matrix for NetFlow and sFlow

Use this ready reckoner to locate features you're interested in and map their availability across platforms and releases.

Table 1: Feature, Release, and Platform Matrix

Feature	Cisco 8000 Series	Other Routing Platforms
BGP Monitoring using IPFIX in MPLS Records	8000, R24.1.1	-
Monitoring Post-QoS Data in NetFlow and IPFIX	8000, R24.1.1	-
BGP Monitoring using IPFIX in MPLS Records	-	NCS 5500, R24.1.1
Simultaneous L2 and L3 Flow Monitoring using IPFIX	-	• ASR 9000, R7.10.1 • NCS 5500, R7.10.1 • NCS 540, R7.10.1 • NCS 560, R7.10.1
sFlow Agent Address Assignment	-	• ASR 9000, R7.10.1 • NCS 5500, R7.10.1

Feature	Cisco 8000 Series	Other Routing Platforms
IPFIX Enablement for SRv6 and Services over SRv6 Core	-	• ASR 9000, R7.10.1 • NCS 560, R7.10.1 • NCS 5500, R7.8.1 • NCS 540, R7.8.1
System Alerts Related to sFlow	8000, R7.5.4	NCS 5500, R7.3.4
Enhanced NetFlow Sampling Rate of 1:2048 (2K)	-	NCS 5500, R7.4.1
MPLS top label type 4 for BGP Labeled Unicast traffic	-	• NCS 5500, R7.4.1 • NCS 540, R7.4.1
Tunnel Encapsulation and Increased sFlow datagram size	-	NCS 5500, R7.3.4
Increased sFlow Sample-Header Size	8000, R7.3.4	-
Ingress sFlow Enhancements	8000, R7.5.4	-
sFlow for L2 Interfaces	8000, R7.5.4	-
Flow Filter on Cisco NC57 Line Cards	-	NCS 5500, R7.2.2
Sampled Flow	8000, R7.2.12	• ASR 9000, R7.5.1 • NCS 5500, R7.5.1 • NCS 540, R7.5.1
IPFIX Flow Record Enhancements for L2 and L3 traffic	8000, R7.2.12	ASR 9000, R7.4.1



CHAPTER 3

What are NetFlow and sFlow Protocols?

NetFlow and sFlow are both network monitoring technologies that provide insights into network traffic and performance.

Table 2: Feature History Table

Feature Name	Release Information	Feature Description
NetFlow and sFlow protocols	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) Network monitoring technologies, such as NetFlow and sFlow provide insights into network traffic performance. NetFlow collects and analyzes network traffic data, allowing you to understand traffic patterns, detect anomalies, and optimize network performance. sFlow, on the other hand, samples packets at the interface level and provides information on the types of traffic and the devices generating it. This feature is now supported on: • 8011-4G24Y4H-I

NetFlow, developed by Cisco, is a protocol that collects and analyzes network traffic data, allowing organizations to understand traffic patterns, detect anomalies, and optimize network performance.

On the other hand, sFlow is a more open and vendor-neutral protocol for monitoring network traffic. It samples packets at the interface level and provides a broader view of network activity, including detailed information on the types of traffic and the devices generating it.

- [Benefits of NetFlow and sFlow, on page 6](#)
- [Key Components of NetFlow and sFlow, on page 6](#)

Benefits of NetFlow and sFlow

Network monitoring and traffic analysis offer insights into traffic, and help you understand network behavior.

Monitoring Network Applications and Use

The data collected through NetFlow or sFlow enables you to view comprehensive, time- and application-based insights into network usage. This data serves as a foundation for strategic network and application resource allocation, offering robust near real-time monitoring capabilities. It can effectively display traffic trends and views based on applications. Additionally, it facilitates proactive identification of issues, streamlined troubleshooting, and swift problem resolution. This information proves invaluable in optimally allocating network resources, as well as identifying and addressing potential security breaches and policy violations.

Network Planning

NetFlow or sFlow can be effectively used to capture data over extended durations, empowering users to monitor and predict network expansion, and plan enhancements such as increased routing devices, ports, or higher-bandwidth interfaces. The data serves as a cornerstone for fine-tuning network planning, including aspects such as peering, backbone upgrades, and routing policy decisions. This approach minimizes overall network operational costs while maximizing performance, capacity, and reliability. The data aids in identifying unwanted WAN traffic, validating bandwidth and Quality of Service (QoS), and facilitating analysis of novel network applications. This wealth of information ultimately contributes to reducing the network operation costs.

Security Analysis

NetFlow or sFlow data plays a pivotal role in promptly detecting and categorizing real-time Denial of Service (DoS) attacks, viruses, and worms. Changes in network patterns reveal anomalies that are distinctly highlighted in the NetFlow data. Furthermore, this data is an invaluable resource for network forensic analysis, enabling a comprehensive understanding and reconstruction of security incidents.

Billing and Accounting

Provides insights into the utilization of resources across a network, and facilitating detailed accounting reports depicting resource usage across diverse network components.

Traffic Engineering

NetFlow and sFlow can gauge the volume of traffic traversing peering or transit points, and assess whether a peering agreement with other service providers is fair and equitable.

Key Components of NetFlow and sFlow

The following NetFlow and sFlow components help you capture, export, collect, analyze, and manage data:

Flow Exporter

The flow exporter, also referred to as an exporter map, functions as a device tasked with collecting data regarding network flows. It transfers the compiled flow records to the designated collector. The exporter is

responsible for inspecting packets, identifying flows, and exporting flow-related data. The exporter map can transmit flow reports to a single destination. A maximum of 8 exporters are permitted per MAP configuration.

Contained within a flow exporter are particulars outlining network specifications and transport layer attributes related to the packets. These packets are exported to the collector through the utilization of the User Datagram Protocol (UDP) transport protocol. In cases where the source interface does not have an assigned IP address, the packet exporter remains inactive.

Flow Monitor

A flow monitor, also referred to as a Monitor map, serves the purpose of facilitating active traffic monitoring on a pre-configured interface. After the flow monitor is committed to an interface, a corresponding flow monitor cache is generated. This cache is used to collect traffic data based on both key and non-key fields outlined within the configured record.

A monitor map contains name references that link to the flow record map and flow exporter map, both of which are committed to an interface. If an exporter map is not applied to the monitor map, the flow records are not exported. In such cases, the aging process adheres to the cache parameters specified in the monitor map.

Furthermore, the option to include extended details such as router-specific elements like nexthop, source and destination mask lengths, and extended gateways attributes, including nexthop, communities, local preference, and AS (source AS, source peer AS, and destination AS path) information.

Flow Sampler

The sampler map specifies the rate at which packets (one out of n packets) are sampled. The sampler map configuration is typically geared for high-speed interfaces to optimize CPU utilization. To achieve this, start by setting the sampling rate after evaluating your network parameters such as traffic rate, number of total flows, cache size, active and inactive timers.

- The maximum supported sampling rate is 1:1, where every packet is processed.
- The minimum supported sampling rate is 1:262,144, indicating that only one out of every 262,144 packets is processed.

Consider these points before applying the sampler map:

- Remove any existing Netflow or sFlow configurations before applying a new sampler map on an interface.
- Use the same sampler map configuration on the sub-interfaces and physical interfaces under a port.



CHAPTER 4

Determine the Monitoring Protocol for Your Network

This section will help you understand the key distinctions between NetFlow and sFlow, equipping you with the necessary insights and knowledge to make an informed decision when it comes to selecting the most suitable monitoring protocol for your network.

- [Key Differences between sFlow and Netflow, on page 9](#)

Key Differences between sFlow and Netflow

This section helps you understand the difference between Netflow and sFlow based on the following factors:

Table 3: Differences between sFlow and Netflow

Factor	Netflow	sFlow
Functionality	Stores the exported metadata aggregated related to IP traffic in the form of a Netflow Record Template. This Netflow Record is saved until the active timer expires, at which point it's exported to the collector.	sFlow is transient and therefore the data is not stored, instead, it's promptly transferred to the collector.
Monitored Flows	Monitors the following traffic: • Layer 2 • Layer 3: MPLS, IPv4, IPv6, SRv6	Monitors the following traffic: • Layer 2 • Layer 3: MPLS, IPv4, IPv6, SRv6 • VLANs
Flow Data Processing	Processes flow data and stores it as flow records.	The flow data is sent to the collector for analysis. Unlike NetFlow, sFlow is not cached and contains fewer flow data fields.
Scalability	Scalability is determined by the number of packets sampled as per the NetFlow sampler map.	Scalability is determined by the number of sFlow packets.

Factor	Netflow	sFlow
Mechanism	Packet aggregation into flows. Netflow is stateful as it stores the Flow record until the active timer expires.	Packets are sampled randomly, while counters are sampled based on time intervals. sFlow is not stateful since the data is not stored; it's directly forwarded to the collector.
Resource usage	Elevated, due to more information provided, resulting in a significant burden on the router.	Typically places lower load on the router.

Summary

Before you identify the best protocol to monitor traffic, you must consider the type of devices that make up your network and the factors that influence the network performance such as latency, scalability and so on.

- Opt for NetFlow when you want to gather extensive data and enhance visibility, though it might introduce a slightly higher latency compared to sFlow. NetFlow offers comprehensive insights into traffic flows. With this information, you can effectively address network issues, identify security risks, strategize network upgrades, and optimize bandwidth utilization.
- Opt for sFlow if you're aiming to oversee traffic in networks with limited bandwidth. sFlow imposes lesser load on network and computing resources compared to NetFlow. Consequently, for small and medium-sized businesses (SMBs) and smaller networks that use less powerful devices, sFlow is more suitable. It can effectively mitigate any performance concerns. sFlow operates with lower resource demands and because of its sampling approach to collect data from a subset of packets. By utilizing this sampled data, you can analyze traffic patterns, enabling the identification of irregularities and the optimization of network performance.

If your network setup accommodates both NetFlow and sFlow, you have the option to leverage both technologies and take advantage of their respective strengths. By identifying specific use cases, you can make a well-informed choice regarding the protocol that meets your network's traffic monitoring needs.



CHAPTER 5

NetFlow Configuration for Traffic Monitoring and Analysis

This page will help you understand the fundamental principles, variations, benefits, and limitations associated with NetFlow. Additionally, it offers guidance on configuring NetFlow.

- [NetFlow Essential Concepts and Terms, on page 11](#)
- [How NetFlow works, on page 12](#)
- [Monitor Traffic Within Your Network, on page 16](#)
- [Interface Types Supported with NetFlow, on page 32](#)
- [NetFlow Guidelines and Limitations, on page 32](#)
- [Comparative Overview of NetFlow Version 9 and Version 10 \(IPFIX\), on page 34](#)
- [NetFlow Version 9 , on page 34](#)
- [IPFIX \(NetFlow version 10\), on page 45](#)
- [Monitoring Post-QoS data in NetFlow and IPFIX, on page 47](#)
- [NetFlow v9 and NetFlow v10 \(IPFIX\), on page 48](#)
- [Revision History, on page 60](#)

NetFlow Essential Concepts and Terms

- **Data source:** Specific locations within the router, such as physical interfaces and VLANs, where traffic measurements can be taken.
- **Flow:** Indicates a collection of IP or MPLS packets traversing the router during a time period. All packets belonging to a particular Flow share common attributes derived from the packet's data
- **Flow record:** Is a set of key and non-key NetFlow field values used to characterize flows in the NetFlow cache. It is generated by examining packet headers, and adding a description of packet details in the NetFlow cache.
- **Exporter:** Positioned within the router that has NetFlow enabled, an Exporter monitors incoming packets, and generates Flows from them. The Exporter transmits information derived from these Flows, encapsulates as Flow Records, to the NetFlow Collector.
- **Collector:** An external device designed to receive Flow Records from one or multiple Exporters. The Collector processes the incoming export packets, and stores the associated Flow record details. Optionally, Flow records can undergo aggregation before storing it onto the hard disk.

- **NetFlow Cache:** The Cache is a segment of memory that stores flow entries prior to their exportation to an external collector. This includes two cache types: the normal cache and the permanent cache.
- **Netflow Analyser:** Is an external device or an application responsible for collecting and scrutinizing flow records to furnish valuable insights.
- **Collector address:** This comprises the IP address and a UDP port number. By default, the designated destination port number is 2055.

How NetFlow works

NetFlow serves as a network monitoring protocol that facilitates the logging of metadata for each flow that traverses the router, both entering or leaving it. This protocol provides comprehensive insights into network flows, including details such as source and destination IP addresses, ports, and packet counts. It's commonly applied for traffic analysis, capacity planning, and network troubleshooting.

Cross AFI BGP NH information elements

Cross AFI BGP NH information elements specifies the next hop IP address for different network layer protocols in BGP routing. These elements ensure

- proper routing across diverse network environments by indicating the appropriate next hop based on the Address Family Identifier (AFI) and
- its Subsequent Address Family Identifier (SAFI).

Table 4: Feature History Table

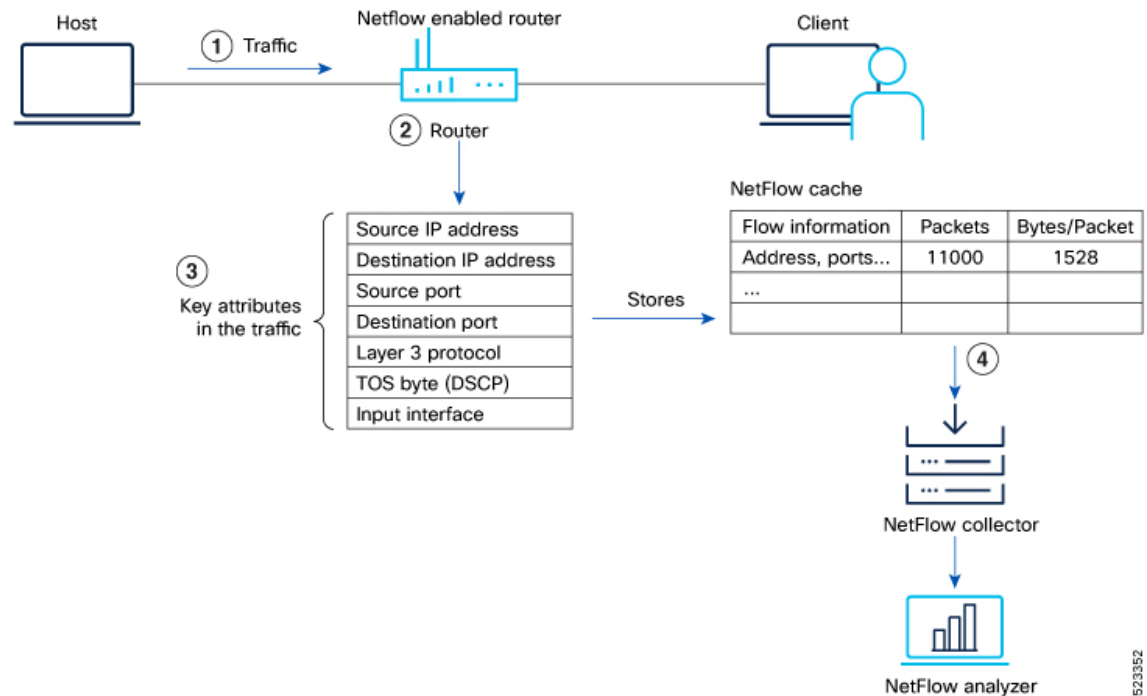
Feature Name	Release Information	Description
Cross AFI BGP NH Information Element	Release 25.1.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). IPv4 or IPv6 flows in BGP can now handle next-hop Information Element (IE) across different address families, such as IPv4 and IPv6. This is particularly useful in scenarios where IPv4 and IPv6 networks need to interoperate. These IEs are added to the existing NetFlow or IPFIX template for <code>record ipv4</code> and all the IPv4 variant record types: • BgpNextHopIPv6Address (IE 63) • IpNextHopIPv6Address (IE 62) • IpNextHopIPv4Address (IE 15) These IEs are added to the existing NetFlow or IPFIX template for <code>record ipv6</code> and all the IPv6 variant record types: • BgpNextHopIPv4Address (IE 18) • IpNextHopIPv6Address (IE 62) • IpNextHopIPv4Address (IE 15) These IEs provide a detailed and structured data that is essential for various network operations and analyses. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH The feature uses the exiting CLI commands. For more information see, IPFIX Enablement for SRv6 and Services over SRv6 Core.

Feature Name	Release Information	Description
Netflow support	Release 25.1.1	Introduced in this release on: Fixed Systems (8700 [ASIC: K100], 8010 [ASIC: A100])(select variants only*) * This feature is now supported on: • 8712-MOD-M • 8011-4G24Y4H-I
Netflow support	Release 24.4.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). NetFlow enhances the system's traffic monitoring capabilities by allowing you to track and analyze IP traffic flows in real-time, offering improved visibility into network utilization and aiding in troubleshooting. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH

Recording of Packet Flows in NetFlow

The packet in NetFlow is recorded as follows:

Figure 1: Packet Flows in NetFlow

**1 Flow Creation****2 Datagram Generation****3 Data Export****4 Analysis and Reporting**

In NetFlow, the focus is on recording and collecting full packet flows in the network traffic data. When NetFlow is configured on the router, the router collects flow data by extracting key field attributes from the packet streams, and generates a flow record. This record, along with accounting information, is stored in the database or NetFlow Cache. The extracted records, once sampled, are exported to one or more NetFlow collectors via the UDP transport layer protocol. This exported data has several purpose: enterprise accounting and ISP billing, and so on.

Here's how NetFlow handles the recording of packet flows:

- Flow Creation**: NetFlow creates flow records by monitoring network traffic passing through the router. As a packet stream traverses a router interface, the packets are collected and an internal header is appended. These packets are dispatched to the line card's CPU, which generate a flow record. The router extracts pertinent header details from the packets and creates cache entries. The packets are subject to a policer, which helps protect the internal control plane. With each subsequent arrival of a packet from the same flow, the cache entry is updated. Flow records persist within the line card's cache until they age out due to timer expiration.

When the expiry of the set timer occurs, the NetFlow is generated. There are timers (two of them) running for flow aging.

- The active timer signifies the maximum allowable duration for a particular cache entry's existence, even if matched by received sampled packets.

- The inactive timer represents the duration without receipt of a sampled packet corresponding to a specific cache entry.
- 2. **Datagram Generation:** The NetFlow agent generates NetFlow datagrams that contain information about the packets. These datagrams include details such as source and destination IP addresses, port numbers, protocol information, and various flow statistics.
- 3. **Data Export:** The NetFlow datagrams are periodically exported from the NetFlow agent to a designated NetFlow collector or analyzer. The export can be done using protocols like UDP or TCP, and the datagrams are typically sent in a structured format like IPFIX or JSON.

A flow record is sent to the NetFlow collector in the following scenarios:

- The flow has been inactive or active for an extended period.
 - The user triggers the export of the flow.
 - The flow concludes, which is particularly relevant when TCP connections are terminated.
4. **Analysis and Reporting:** Upon receiving the NetFlow data, the NetFlow collector or analyzer processes and analyzes the information. It aggregates the sampled data to provide statistical insights into network traffic, including top talkers, protocol distribution, traffic patterns, and other metrics.

Monitor Traffic Within Your Network

NetFlow extends its support to IPv4, IPv6, MPLS, BGP, SRv6, and GTP-U flow types, providing the capacity to monitor a diverse range of packet information.

Monitor IP Traffic

NetFlow can be used to collect traffic data for both IPv4 and IPv6 networks. The data collected includes information such as source and destination IP addresses, protocol types, port numbers, and bandwidth usage. This data can be used to identify network trends, detect security threats, and optimize network performance.

Key IP traffic attributes monitored:

- Source and Destination IP Addresses
- Source and Destination MAC Addresses
- Source and Destination Ports for TCP/User Datagram Protocol (UDP) ports
- Differentiated Services Code Point (DSCP)
- Layer 3 Protocol
- Type of Service (ToS) Byte
- Traffic receiving Interface
- Complete IPv4 Header fields, including IP-ID and TTL, among others
- Counts for Packets and Bytes
- Full Spectrum of IPv6 Header fields, encompassing Flow Label and Option Header, among others

- Flow timestamps

Monitor MPLS Traffic

NetFlow can be used to collect traffic data for MPLS traffic in a networks. NetFlow provides detailed visibility into MPLS traffic, allowing you to identify anomalies, detect cyber threats, and respond quickly to potential security incidents

Key MPLS traffic attribute monitored:

- MPLS Labels

Monitor BGP Traffic

NetFlow can be used to monitor BGP traffic in your network. NetFlow can capture BGP packets and provide information on their frequency, direction, and content. This information can be used to identify potential security threats and monitor BGP router behavior. BGP is used by network routers to exchange routing information and establish optimal paths for data to travel through a network. By monitoring BGP communication with NetFlow, you can gain valuable insights into their network's performance and ensure that their routing strategies are effective.

Key BGP traffic attributes monitored:

- Next-hop address
- Source autonomous system (AS) number
- Destination autonomous system (AS) number
- Source prefix mask
- Destination prefix mask
- BGP Next Hop
- BGP Policy Accounting traffic index

Monitor SRv6 Traffic

During the transition from conventional IP/MPLS networks to SRv6-based networks, the requirement for information elements specific to SRv6 traffic flow in the core arises. To address this requirement, we have introduced support for SRv6 monitoring using Netflow.

Starting from IOS-XR software release 7.10.1, you can monitor the performance of SRv6-based core networks using IPFIX.

Restriction and Limitation

- SRv6 traffic monitoring using IPFIX is supported only on P-nodes; decapsulation nodes are not supported.

Monitor GTP-U Traffic in 5G Network

Cisco 8000 routers introduces the capability to monitor the performance of GTP-U traffic in 5G networks. This feature utilizes Netflow and IPFIX to collect and analyze traffic data, offering valuable insights into network performance and facilitating effective management of 5G network traffic.

Starting from IOS-XR software release 24.2.1, three new GTP-U related information elements can be gathered in Netflow and IPFIX records for both IPv4 and IPv6 traffic. This advancement allows administrators to optimize the performance and security of their 5G networks.

The newly introduced information elements are as follows:

IE Field	IE Number
GTP_TEID	507
GTP_QFI	509
GTP_SESS_DIR	510

IE number, or Information Element Number, is a unique identifier assigned to specific elements within network communication protocols, facilitating standardized interpretation and management. For more information, refer IP Flow Information Export (IPFIX) Entities.

Benefits of GTP-U Traffic Monitoring

The following are some of the key benefits of enabling GTP-U traffic monitoring on your router.

- **Monitor Network Slicing:** 5G network slicing enables the creation of dedicated virtual networks with specific functionalities. By exporting GTP traffic records, you can conduct detailed analysis of the traffic within each slice, ensuring optimal performance and resource allocation.
- **Flexible Deployment:** GTP-U monitoring can be implemented on any network node where the outermost traffic encapsulation utilizes the GTP protocol. This capability can be activated to monitor traffic at various strategic points across the network infrastructure.
- **IPv6 Support for 5G Deployments:** With the expansion of 5G networks, there's an increasing use of IPv6, especially in scenarios where 5G base stations (gNodeBs) connect to User Plane Functions (UPFs) using IPv6. This feature ensures that flow records for such IPv6 GTP-U traffic can be captured and exported effectively.

GTP-U Traffic Record Templates

This section provides you with all the record template options available for monitoring GTP-U traffic.

IPv4-GTP-IPv4 Record

This record captures GTP-U traffic details between IPv4 interfaces, essential for monitoring and optimizing IPv4 5G network performance.

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
46	256	V9_ETH_TYPE	2	256	V9_ETH_TYPE	2

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
47	243	V9_DOT1Q_VLAN_ID	2	243	V9_DOT1Q_VLAN_ID	2
48	245	V9_DOT1Q_CUST_VLAN_ID	2	245	V9_DOT1Q_CUST_VLAN_ID	2
49	244	V9_DOT1Q_PRIORITY	1	244	V9_DOT1Q_PRIORITY	1
50	198	IN_BYTES_DELTA	8	444	V9_AS_PATH	128
1	2	V9_IN_PKTS	8	2	V9_IN_PKTS	4
2	1	V9_IN_BYTES	8	1	V9_IN_BYTES	4
3	10	V9_INPUT_SNMP	4	10	V9_INPUT_SNMP	4
4	14	V9_OUTPUT_SNMP	4	14	V9_OUTPUT_SNMP	4
5	22	V9_FIRST_SWITCHED	4	22	V9_FIRST_SWITCHED	4
6	21	V9_LAST_SWITCHED	4	21	V9_LAST_SWITCHED	4
7	89	V9_FORWARDING_STATUS	4	89	V9_FORWARDING_STATUS	1
8	61	V9_DIRECTION	1	61	V9_DIRECTION	1
9	302	SELECTOR_ID	4	48	V9_FLOW_SAMPLER_ID	2
10	234	V9_VRF_ID_INPUT	4	234	V9_VRF_ID_INPUT	4
11	235	V9_VRF_ID_OUTPUT	4	235	V9_VRF_ID_OUTPUT	4
12	55	V9_POST_QOS_TOS	1	55	V9_POST_QOS_TOS	1
13	8	V9_IPV4SRCADDR	4	8	V9_IPV4SRCADDR	4
14	12	V9_IPV4DSTADDR	4	12	V9_IPV4DSTADDR	4
15	7	V9_SRC_PORT	2	7	V9_SRC_PORT	2
16	11	V9_DST_PORT	2	11	V9_DST_PORT	2
17	9	V9_SRC_MASK	1	9	V9_SRC_MASK	1
18	13	V9_DST_MASK	1	13	V9_DST_MASK	1
19	4	V9_PROT	1	4	V9_PROT	1
20	6	V9_TCP_FLAGS	2	6	V9_TCP_FLAGS	1
21	5	V9_TOS	1	5	V9_TOS	1
22	52	V9_MIN_TTL	1	52	V9_MIN_TTL	1
23	53	V9_MAX_TTL	1	53	V9_MAX_TTL	1
24	54	V9_IP_IDENT	4	54	V9_IP_IDENT	4
25	197	IPFIX_FRAG_FLAGS	1	197	IPFIX_FRAG_FLAGS	1
26	88	V9_FRAGMENT_OFFSET	2	88	V9_FRAGMENT_OFFSET	2

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
27	184	IPFIX_TCP_SEQ_NUM	4	184	IPFIX_TCP_SEQ_NUM	4
28	25	V9_MIN_PKT_LEN	8	25	V9_MIN_PKT_LEN	8
29	26	V9_MAX_PKT_LEN	8	26	V9_MAX_PKT_LEN	8
30	503	IPFIX_L4_CHECKSUM	2	503	IPFIX_L4_CHECKSUM	2
31	504	IPFIX_ICMP_8_BYTES	8	504	IPFIX_ICMP_8_BYTES	8
32	507	GTP_TEID	4	507	GTP_TEID	4
33	509	GTP_QFI	1	509	GTP_QFI	1
34	510	GTP_SESS_DIR	1	510	GTP_SESS_DIR	1
35	8	V9_IPV4SRCADDR	4	8	V9_IPV4SRCADDR	4
36	12	V9_IPV4DSTADDR	4	12	V9_IPV4DSTADDR	4
37	5	V9_TOS	1	5	V9_TOS	1
38	16	V9_SRC_AS	4	16	V9_SRC_AS	4
39	17	V9_DST_AS	4	17	V9_DST_AS	4
40	18	V9_BGP_IPV4_NEXT_HOP	4	18	V9_BGP_IPV4_NEXT_HOP	4
41	63	V9_BGP_IPV6_NEXT_HOP	16	63	V9_BGP_IPV6_NEXT_HOP	16
42	15	V9_IPV4_NEXT_HOP	4	15	V9_IPV4_NEXT_HOP	4
43	62	V9_IPV6_NEXT_HOP	16	62	V9_IPV6_NEXT_HOP	16
44	56	V9_IN_SRC_MAC	6	56	V9_IN_SRC_MAC	6
45	80	V9_IN_DST_MAC	6	80	V9_IN_DST_MAC	6
46	256	V9_ETH_TYPE	2	256	V9_ETH_TYPE	2
47	243	V9_DOT1Q_VLAN_ID	2	243	V9_DOT1Q_VLAN_ID	2
48	245	V9_DOT1Q_CUST_VLAN_ID	2	245	V9_DOT1Q_CUST_VLAN_ID	2
49	244	V9_DOT1Q_PRIORITY	1	244	V9_DOT1Q_PRIORITY	1
50	198	IN_BYTES_DELTA	8	444	V9_AS_PATH	128
51				445	V9_STD_COMM	128

IPv4-GTP-IPv6 Record

This record monitors GTP-U traffic that starts in an IPv4 network and transitions into an IPv6 network, aiding in cross-network compatibility analysis.

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
1	2	V9_IN_PKTS	8	2	V9_IN_PKTS	4
2	1	V9_IN_BYTES	8	1	V9_IN_BYTES	4
3	10	V9_INPUT_SNMP	4	10	V9_INPUT_SNMP	4
4	14	V9_OUTPUT_SNMP	4	14	V9_OUTPUT_SNMP	4
5	22	V9_FIRST_SWITCHED	4	22	V9_FIRST_SWITCHED	4
6	21	V9_LAST_SWITCHED	4	21	V9_LAST_SWITCHED	4
7	89	V9_FORWARDING_STATUS	4	89	V9_FORWARDING_STATUS	1
8	61	V9_DIRECTION	1	61	V9_DIRECTION	1
9	302	SELECTOR_ID	4	48	V9_FLOW_SAMPLER_ID	2
10	234	V9_VRF_ID_INPUT	4	234	V9_VRF_ID_INPUT	4
11	235	V9_VRF_ID_OUTPUT	4	235	V9_VRF_ID_OUTPUT	4
12	55	V9_POST_QOS_TOS	1	55	V9_POST_QOS_TOS	1
13	27	V9_IPV6_SRC_ADDR	16	27	V9_IPV6_SRC_ADDR	16
14	28	V9_IPV6_DST_ADDR	16	28	V9_IPV6_DST_ADDR	16
15	31	V9_FLOW_LABEL	4	31	V9_FLOW_LABEL	3
16	64	V9_IPV6_OPTION_HEADERS	4	64	V9_IPV6_OPTION_HEADERS	4
17	7	V9_SRC_PORT	2	7	V9_SRC_PORT	2
18	11	V9_DST_PORT	2	11	V9_DST_PORT	2
19	30	V9_IPV6_DST_MASK	1	30	V9_IPV6_DST_MASK	1
20	29	V9_IPV6_SRC_MASK	1	29	V9_IPV6_SRC_MASK	1
21	4	V9_PROT	1	4	V9_PROT	1
22	6	V9_TCP_FLAGS	2	6	V9_TCP_FLAGS	1
23	5	V9_TOS	1	5	V9_TOS	1
24	52	V9_MIN_TTL	1	52	V9_MIN_TTL	1
25	53	V9_MAX_TTL	1	53	V9_MAX_TTL	1
26	54	V9_IP_IDENT	4	54	V9_IP_IDENT	4
27	197	IPFIX_FRAG_FLAGS	1	197	IPFIX_FRAG_FLAGS	1
28	88	V9_FRAGMENT_OFFSET	2	88	V9_FRAGMENT_OFFSET	2
29	184	IPFIX_TCP_SEQ_NUM	4	184	IPFIX_TCP_SEQ_NUM	4
30	25	V9_MIN_PKT_LEN	8	25	V9_MIN_PKT_LEN	8

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
31	26	V9_MAX_PKT_LEN	8	26	V9_MAX_PKT_LEN	8
32	503	IPFIX_L4_CHECKSUM	2	503	IPFIX_L4_CHECKSUM	2
33	504	IPFIX_ICMP_8_BYTES	8	504	IPFIX_ICMP_8_BYTES	8
34	507	GTP_TEID	4	507	GTP_TEID	4
35	509	GTP_QFI	1	509	GTP_QFI	1
36	510	GTP_SESS_DIR	1	510	GTP_SESS_DIR	1
37	8	V9_IPV4SRCADDR	4	8	V9_IPV4SRCADDR	4
38	12	V9_IPV4DSTADDR	4	12	V9_IPV4DSTADDR	4
39	5	V9_TOS	1	5	V9_TOS	1
40	16	V9_SRC_AS	4	16	V9_SRC_AS	4
41	17	V9_DST_AS	4	17	V9_DST_AS	4
42	18	V9_BGP_IPV4_NEXT_HOP	4	18	V9_BGP_IPV4_NEXT_HOP	4
43	63	V9_BGP_IPV6_NEXT_HOP	16	63	V9_BGP_IPV6_NEXT_HOP	16
44	15	V9_IPV4_NEXT_HOP	4	15	V9_IPV4_NEXT_HOP	4
45	62	V9_IPV6_NEXT_HOP	16	62	V9_IPV6_NEXT_HOP	16
46	56	V9_IN_SRC_MAC	6	56	V9_IN_SRC_MAC	6
47	80	V9_IN_DST_MAC	6	80	V9_IN_DST_MAC	6
48	256	V9_ETH_TYPE	2	256	V9_ETH_TYPE	2
49	243	V9_DOT1Q_VLAN_ID	2	243	V9_DOT1Q_VLAN_ID	2
50	245	V9_DOT1Q_CUST_VLAN_ID	2	245	V9_DOT1Q_CUST_VLAN_ID	2
51	244	V9_DOT1Q_PRIORITY	1	244	V9_DOT1Q_PRIORITY	1
52	198	IN_BYTES_DELTA	8	444	V9_AS_PATH	128
53				445	V9_STD_COMM	128

IPv6-GTP-IPv4 Record

This record monitors GTP-U traffic moving from an IPv6 network to an IPv4 network, ensuring seamless data flow across different network types.

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
1	2	V9_IN_PKTS	8	2	V9_IN_PKTS	4

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
2	1	V9_IN_BYTES	8	1	V9_IN_BYTES	4
3	10	V9_INPUT_SNMP	4	10	V9_INPUT_SNMP	4
4	14	V9_OUTPUT_SNMP	4	14	V9_OUTPUT_SNMP	4
5	21	V9_LAST_SWITCHED	4	21	V9_LAST_SWITCHED	4
6	22	V9_FIRST_SWITCHED	4	22	V9_FIRST_SWITCHED	4
7	89	V9_FORWARDING_STATUS	4	89	V9_FORWARDING_STATUS	1
8	61	V9_DIRECTION	1	61	V9_DIRECTION	1
9	302	SELECTOR_ID	4	48	V9_FLOW_SAMPLER_ID	2
10	234	V9_VRF_ID_INPUT	4	234	V9_VRF_ID_INPUT	4
11	235	V9_VRF_ID_OUTPUT	4	235	V9_VRF_ID_OUTPUT	4
12	55	V9_POST_QOS_TOS	1	55	V9_POST_QOS_TOS	1
13	8	V9_IPV4SRCADDR	4	8	V9_IPV4SRC4ADDR	4
14	12	V9_IPV4DSTADDR	4	12	V9_IPV4DSTADDR	4
15	7	V9_SRC_PORT	2	7	V9_SRC_PORT	2
16	11	V9_DST_PORT	2	11	V9_DST_PORT	2
17	9	V9_SRC_MASK	1	9	V9_SRRC_MASK	1
18	13	V9_DST_MASK	1	13	V9_DST_MASK	1
19	4	V9_PROT	1	4	V9_PROT	1
20	6	V9_TCP_FLAGS	2	6	V9_TCP_FLAGS	1
21	5	V9_TOS	1	5	V9_TOS	1
22	52	V9_MIN_TTL	1	52	V9_MIN_TTL	1
23	53	V9_MAX_TTL	1	53	V9_MAX_TTL	1
24	54	V9_IP_IDENT	4	54	V9_IP_IDENT	4
25	197	IPFIX_FRAG_FLAGS	1	197	IPFIX_FRAG_FLAGS	1
26	88	V9_FRAGMENT_OFFSET	2	88	V9_FRAGMENT_OFFSET	2
27	184	IPFIX_TCP_SEQ_NUM	4	184	IPFIX_TCP_SEQ_NUM	4
28	25	V9_MIN_PKT_LEN	8	25	V9_MIN_PKT_LEN	8
29	26	V9_MAX_PKT_LEN	8	26	V9_MAX_PKT_LEN	8
30	503	IPFIX_L4_CHECKSUM	2	503	IPFIX_L4_CHECKSUM	2
31	504	IPFIX_ICMP_8_BYTES	8	504	IPFIX_ICMP_8_BYTES	8

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
32	507	GTP_TEID	4	507	GTP_TEID	4
33	509	GTP_QFI	1	509	GTP_QFI	1
34	510	GTP_SESS_DIR	1	510	GTP_SESS_DIR	1
35	27	V9_IPV6_SRC_ADDR	16	27	V9_IPV6_SRC_ADDR	16
36	28	V9_IPV6_DST_ADDR	16	28	V9_IPV6_DST_ADDR	16
37	5	V9_TOS	1	5	V9_TOS	1
38	31	V9_FLOW_LABEL	4	31	V9_FLOW_LABEL	3
39	16	V9_SRC_AS	4	16	V9_SRC_AS	4
40	17	V9_DST_AS	4	17	V9_DST_AS	4
41	18	V9_BGP_IPV4_NEXT_HOP	4	18	V9_BGP_IPV4_NEXT_HOP	4
42	63	V9_BGP_IPV6_NEXT_HOP	16	63	V9_BGP_IPV6_NEXT_HOP	16
43	15	V9_IPV4_NEXT_HOP	4	15	V9_IPV4_NEXT_HOP	4
44	62	V9_IPV6_NEXT_HOP	16	62	V9_IPV6_NEXT_HOP	16
45	56	V9_IN_SRC_MAC	6	56	V9_IN_SRC_MAC	6
46	80	V9_IN_DST_MAC	6	80	V9_IN_DST_MAC	6
47	256	V9_ETH_TYPE	2	256	V9_ETH_TYPE	2
48	243	V9_DOT1Q_VLAN_ID	2	243	V9_DOT1Q_VLAN_ID	2
49	245	V9_DOT1Q_CUST_VLAN_ID	2	245	V9_DOT1Q_CUST_VLAN_ID	2
50	244	V9_DOT1Q_PRIORITY	1	244	V9_DOT1Q_PRIORITY	1
51	198	IN_BYTES_DELTA	8	444	V9_AS_PATH	128
52				445	V9_STD_COMM	128

IPv6-GTP-IPv6 Record

This record provides insights into GTP-U traffic within IPv6 networks, crucial for maintaining the integrity and efficiency of modern 5G infrastructures.

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
1	2	V9_IN_PKTS	8	2	V9_IN_PKTS	4
2	1	V9_IN_BYTES	8	1	V9_IN_BYTES	4
3	10	V9_INPUT_SNMP	4	10	V9_INPUT_SNMP	4

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
4	14	V9_OUTPUT_SNMP	4	14	V9_OUTPUT_SNMP	4
5	21	V9_LAST_SWITCHED	4	21	V9_LAST_SWITCHED	4
6	22	V9_FIRST_SWITCHED	4	22	V9_FIRST_SWITCHED	4
7	89	V9_FORWARDING_STATUS	4	89	V9_FORWARDING_STATUS	1
8	61	V9_DIRECTION	1	61	V9_DIRECTION	1
9	302	SELECTOR_ID	4	48	V9_FLOW_SAMPLER_ID	2
10	234	V9_VRF_ID_INPUT	4	234	V9_VRF_ID_INPUT	4
11	235	V9_VRF_ID_OUTPUT	4	235	V9_VRF_ID_OUTPUT	4
12	55	V9_POST_QOS_TOS	1	55	V9_POS_QOS_TOS	1
13	27	V9_IPV6_SRC_ADDR	16	27	V9_IPV6_SRC_ADDR	16
14	28	V9_IPV6_DST_ADDR	16	28	V9_IPV6_DST_ADDR	16
15	31	V9_FLOW_LABEL	4	31	V9_FLOW_LABEL	3
16	64	V9_IPV6_OPTION_HEADERS	4	64	V9_IPV6_OPTION_HEADERS	4
17	7	V9_SRC_PORT	2	7	V9_SRC_PORT	2
18	11	V9_DST_PORT	2	11	V9_DST_PORT	2
19	30	V9_IPV6_DST_MASK	1	30	V9_IPV6_DST_MASK	1
20	29	V9_IPV6_SRC_MASK	1	29	V9_IPV6_SRC_MASK	1
21	4	V9_PROT	1	4	V9_PROT	1
22	6	V9_TCP_FLAGS	2	6	V9_TCP_FLAGS	1
23	5	V9_TOS	1	5	V9_TOS	1
24	52	V9_MIN_TTL	1	52	V9_MIN_TTL	1
25	53	V9_MAX_TTL	1	53	V9_MAX_TTL	1
26	54	V9_IP_IDENT	4	54	V9_IP_IDENT	4
27	197	IPFIX_FRAG_FLAGS	1	197	IPFIX_FRAG_FLAGS	1
28	88	V9_FRAGMENT_OFFSET	2	88	V9_FRAGMENT_OFFSET	2
29	184	IPFIX_TCP_SEQ_NUM	4	184	IPFIX_TCP_SEQ_NUM	4
30	25	V9_MIN_PKT_LEN	8	25	V9_MIN_PKT_LEN	8
31	26	V9_MAX_PKT_LEN	8	26	V9_MAX_PKT_LEN	8
32	503	IPFIX_L4_CHECKSUM	2	503	IPFIX_L4_CHECKSUM	2
33	504	IPFIX_ICMP_8_BYTES	8	504	IPFIX_ICMP_8_BYTES	8

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
34	507	GTP_TEID	4	507	GTP_TEID	4
35	509	GTP_QFI	1	509	GTP_QFI	1
36	510	GTP_SESS_DIR	1	510	GTP_SESS_DIR	1
37	27	V9_IPV6_SRC_ADDR	16	27	V9_IPV6_SRC_ADDR	16
38	28	V9_IPV6_DST_ADDR	16	28	V9_IPV6_DST_ADDR	16
39	5	V9_TOS	1	5	V9_TOS	1
40	31	V9_FLOW_LABEL	4	31	V9_FLOW_LABEL	3
41	16	V9_SRC_AS	4	16	V9_SRC_AS	4
42	17	V9_DST_AS	4	17	V9_DST_AS	4
43	18	V9_BGP_IPV4_NEXT_HOP	4	18	V9_BGP_IPV4_NEXT_HOP	4
44	63	V9_BGP_IPV6_NEXT_HOP	16	63	V9_BGP_IPV6_NEXT_HOP	16
45	15	V9_IPV4_NEXT_HOP	4	15	V9_IPV4_NEXT_HOP	4
46	62	V9_IPV6_NEXT_HOP	16	62	V9_IPV6_NEXT_HOP	16
47	56	V9_IN_SRC_MAC	6	56	V9_IN_SRC_MAC	6
48	80	V9_IN_DST_MAC	6	80	V9_IN_DST_MAC	6
49	256	V9_ETH_TYPE	2	256	V9_ETH_TYPE	2
50	243	V9_DOT1Q_VLAN_ID	2	243	V9_DOT1Q_VLAN_ID	2
51	245	V9_DOT1Q_CUST_VLAN_ID	2	245	V9_DOT1Q_CUST_VLAN_ID	2
52	244	V9_DOT1Q_PRIORITY	1	244	V9_DOT1Q_PRIORITY	1
53	198	IN_BYTES_DELTA	8	444	V9_AS_PATH	128
54				445	V9_STD_COMM	128

Extended Template Records

IPv4 Peering Extended Record

This record extends monitoring capabilities to include detailed peering information for IPv4 traffic, enhancing traffic management and security measures.

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
1	2	V9_IN_PKTS	8	2	V9_IN_PKTS	4
2	1	V9_IN_BYTES	8	1	V9_IN_BYTES	4

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
3	8	V9_IPV4SRCADDR	4	8	V9_IPV4SRCADDR	4
4	12	V9_IPV4DSTADDR	4	12	V9_IPV4DSTADDR	4
5	10	V9_INPUT_SNMP	4	10	V9_INPUT_SNMP	4
6	14	V9_OUTPUT_SNMP	4	14	V9_OUTPUT_SNMP	4
7	22	V9_FIRST_SWITCHED	4	22	V9_FIRST_SWITCHED	4
8	21	V9_LAST_SWITCHED	4	21	V9_LAST_SWITCHED	4
9	7	V9_SRC_PORT	2	7	V9_SRC_PORT	2
10	11	V9_DST_PORT	2	11	V9_DST_PORT	2
11	16	V9_SRC_AS	4	16	V9_SRC_AS	4
12	17	V9_DST_AS	4	17	V9_DST_AS	4
13	18	V9_BGP_IPV4_NEXT_HOP	4	18	V9_BGP_IPV6_NEXT_HOP	4
14	63	V9_BGP_IPV6_NEXT_HOP	16	63	V9_BGP_IPV6_NEXT_HOP	16
15	15	V9_IPV4_NEXT_HOP	4	15	V9_IPV4_NEXT_HOP	4
16	62	V9_IPV6_NEXT_HOP	16	62	V9_IPV6_NEXT_HOP	16
17	9	V9_SRC_MASK	1	9	V9_SRC_MASK	1
18	13	V9_DST_MASK	1	13	V9_DST_MASK	1
19	4	V9_PROT	1	4	V9_PROT	1
20	6	V9_TCP_FLAGS	2	6	V9_TCP_FLAGS	1
21	5	V9_TOS	1	5	V9_TOS	1
22	55	V9_POST_QOS_TOS	1	55	V9_POST_QOS_TOS	1
23	61	V9_DIRECTION	1	61	V9_DIRECTION	1
24	89	V9_FORWARDING_STATUS	4	89	V9_FORWARDING_STATUS	1
25	302	SELECTOR_ID	4	48	V9_FLOW_SAMPLER_ID	2
26	234	V9_VRF_ID_INPUT	4	234	V9_VRF_ID_INPUT	4
27	235	V9_VRF_ID_OUTPUT	4	235	V9_VRF_ID_OUTPUT	4
28	52	V9_MIN_TTL	1	52	V9_MIN_TTL	1
29	53	V9_MAX_TTL	1	53	V9_MAX_TTL	1
30	54	V9_IP_IDENT	4	54	V9_IP_IDENT	4
31	197	IPFIX_FRAG_FLAGS	1	197	IPFIX_FRAG_FLAGS	1
32	88	V9_FRAGMENT_OFFSET	2	88	V9_FRAGMENT_OFFSET	2

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
33	184	IPFIX_TCP_SEQ_NUM	4	184	IPFIX_TCP_SEQ_NUM	4
34	25	V9_MIN_PKT_LEN	8	25	V9_MIN_PKT_LEN	8
35	26	V9_MAX_PKT_LEN	8	26	V9_MAX_PKT_LEN	8
36	503	IPFIX_L4_CHECKSUM	2	503	IPFIX_L4_CHECKSUM	2
37	504	IPFIX_ICMP_8_BYTES	8	504	IPFIX_ICMP_8_BYTES	8
38	56	V9_IN_SRC_MAC	6	56	V9_IN_SRC_MAC	6
39	80	V9_IN_DST_MAC	6	80	V9_IN_DST_MAC	6
40	256	V9_ETH_TYPE	2	256	V9_ETH_TYPE	2
41	243	V9_DOT1Q_VLAN_ID	2	243	V9_DOT1Q_VLAN_ID	2
42	245	V9_DOT1Q_CUST_VLAN_ID	2	245	V9_DOT1Q_CUST_VLAN_ID	2
43	244	V9_DOT1Q_PRIORITY	1	244	V9_DOT1Q_PRIORITY	1
44	198	IN_BYTES_DELTA	8	444	V9_AS_PATH	128
45				445	V9_STD_COMM	128

IPv6 Peering Extended Record

This record offers comprehensive peering data for IPv6 traffic, supporting advanced traffic analysis and network optimization strategies.

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
1	2	V9_IN_PKTS	8	2	V9_IN_PKTS	4
2	1	V9_IN_BYTES	8	1	V9_IN_BYTES	4
3	27	V9_IPV6_SRC_ADDR	16	27	V9_IPV6_SRC_ADDR	16
4	28	V9_IPV6_DST_ADDR	16	28	V9_IPV6_DST_ADDR	16
5	10	V9_INPUT_SNMP	4	10	V9_INPUT_SNMP	4
6	14	V9_OUTPUT_SNMP	4	14	V9_OUTPUT_SNMP	4
7	22	V9_FIRST_SWITCHED	4	22	V9_FIRST_SWITCHED	4
8	21	V9_LAST_SWITCHED	4	21	V9_LAST_SWITCHED	4
9	31	V9_FLOW_LABEL	4	31	V9_FLOW_LABEL	3
10	64	V9_IPV6_OPTION_HEADERS	4	64	V9_IPV6_OPTION_HEADERS	4
11	7	V9_SRC_PORT	2	7	V9_SRC_PORT	2

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
12	11	V9_DST_PORT	2	11	V9_DST_PORT	2
13	16	V9_SRC_AS	4	16	V9_SRC_AS	4
14	17	V9_DST_AS	4	17	V9_DST_AS	4
15	18	V9_BGP_IPV4_NEXT_HOP	4	18	V9_BGP_IPV6_NEXT_HOP	4
16	63	V9_BGP_IPV6_NEXT_HOP	16	63	V9_BGP_IPV6_NEXT_HOP	16
17	15	V9_IPV4_NEXT_HOP	4	15	V9_IPV4_NEXT_HOP	4
18	62	V9_IPV6_NEXT_HOP	16	62	V9_IPV6_NEXT_HOP	16
19	30	V9_IPV6_DST_MASK	1	30	V9_IPV6_DST_MASK	1
20	29	V9_IPV6_SRC_MASK	1	29	V9_IPV6_SRC_MASK	1
21	4	V9_PROT	1	4	V9_PROT	1
22	6	V9_TCP_FLAGS	2	6	V9_TCP_FLAGS	1
23	5	V9_TOS	1	5	V9_TOS	1
24	55	V9_POST_QOS_TOS	1	55	V9_POST_QOS_TOS	1
25	61	V9_DIRECTION	1	61	V9_DIRECTION	1
26	89	V9_FORWARDING_STATUS	4	89	V9_FORWARDING_STATUS	1
27	302	SELECTOR_ID	4	48	V9_FLOW_SAMPLER_ID	2
28	234	V9_VRF_ID_INPUT	4	234	V9_VRF_ID_INPUT	4
29	235	V9_VRF_ID_OUTPUT	4	235	V9_VRF_ID_OUTPUT	4
30	52	V9_MIN_TTL	1	52	V9_MIN_TTL	1
31	53	V9_MAX_TTL	1	53	V9_MAX_TTL	1
32	54	V9_IP_IDENT	4	54	V9_IP_IDENT	4
33	197	IPFIX_FRAG_FLAGS	1	197	IPFIX_FRAG_FLAGS	1
34	88	V9_FRAGMENT_OFFSET	2	88	V9_FRAGMENT_OFFSET	2
35	184	IPFIX_TCP_SEQ_NUM	4	184	IPFIX_TCP_SEQ_NUM	4
36	25	V9_MIN_PKT_LEN	8	25	V9_MIN_PKT_LEN	8
37	26	V9_MAX_PKT_LEN	8	26	V9_MAX_PKT_LEN	8
38	503	IPFIX_L4_CHECKSUM	2	503	IPFIX_L4_CHECKSUM	2
39	504	IPFIX_ICMP_8_BYTES	8	504	IPFIX_ICMP_8_BYTES	8
40	56	V9_IN_SRC_MAC	6	56	V9_IN_SRC_MAC	6
41	80	V9_IN_DST_MAC	6	80	V9_IN_DST_MAC	6

S.No	IPFIX			NetFlow V9		
	IE #	Field	Size (Bytes)	IE #	Field	Size (Bytes)
42	256	V9_ETH_TYPE	2	256	V9_ETH_TYPE	2
43	243	V9_DOT1Q_VLAN_ID	2	243	V9_DOT1Q_VLAN_ID	2
44	245	V9_DOT1Q_CUST_VLAN_ID	2	245	V9_DOT1Q_CUST_VLAN_ID	2
45	244	V9_DOT1Q_PRIORITY	1	244	V9_DOT1Q_PRIORITY	1
46	198	IN_BYTES_DELTA	8	444	V9_AS_PATH	128
47				445	V9_STD_COMM	128

Monitor PWHE Interface Traffic

Table 5: Feature History Table

Feature Name	Release Information	Feature Description
Monitor PWHE Interface Traffic	Release 25.1.1	Introduced in this release on: Fixed Systems (8700 [ASIC:K100])(select variants only*) * This feature is now supported on Cisco 8712-MOD-M routers.

Feature Name	Release Information	Feature Description
Monitor PWHE Interface Traffic	Release 24.3.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) NetFlow is now supported on Pseudowire Headend (PWHE) interfaces, enhancing traffic congestion monitoring and network operation efficiency. This feature enables monitoring of IP traffic transmitted through PWHE interfaces, which facilitates the termination and encapsulation of legacy non-IP traffic into IP packets. *This feature is supported on: • 8212-48FH-M • 8711-32FH-M • 88-LC1-36EH • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM

NetFlow now effectively monitors and manages traffic on Pseudowire Headend (PWHE) interfaces. These interfaces are essential for monitoring legacy services within modern IP/MPLS networks. A PWHE router, located at the network's edge, is designed to terminate pseudowire connections from various legacy non-IP traffic sources. It then encapsulates this traffic into native IP packets. This enhancement provides network administrators with comprehensive visibility, enabling them to make well-informed decisions for maintaining optimal network performance.

Before Cisco IOS XR Release 24.3.1, NetFlow was not supported on PWHE interfaces. The latest update extends NetFlow's functionality to include both IPv4 and IPv6 traffic flows on PWHE interfaces.

Benefits of PWHE Traffic Monitoring

The following are some of the key benefits of monitoring traffic on PWHE interface.

- Detailed traffic insights: NetFlow on PWHE interfaces provides visibility into traffic types, volumes, and patterns, essential for network analysis.
- Enhanced network management: Enables comprehensive traffic analysis for accurate accounting, billing and effective troubleshooting.
- Comprehensive monitoring: Offers both ingress and egress traffic monitoring on PWHE main and sub-interfaces for a complete network performance view.

Interface Types Supported with NetFlow

- Physical main interfaces
- L3 interfaces
- L3 subinterfaces
- L2 interfaces
- Bundle interfaces
- Bundle sub-interfaces
- PW-Ether interfaces
- BVI interfaces

NetFlow Guidelines and Limitations

General

- NetFlow is supported in the ingress direction for all routers.
- NetFlow supports export format Version 9 and IPFIX.
- The configuration of the sampler rate as 1 out of 1 is supported.
- You can configure only one active sampler per system.
- Netflow does not cache for MPLS decap and GRE decap nodes when these are configured on sub-interfaces; cache support is available only on main interfaces.
- Netflow record collection is not supported on SRv6 decap node.
- Ingress L2 netflow or IPFIX is not supported.

Features not Supported

- Egress NetFlow and Internet Protocol Flow Information Export (IPFIX) are not supported.
- The full Packet Capture (FPC) feature is not supported.
- Destination-based NetFlow accounting is not supported.
- NetFlow filtering using ACL is not supported.
- The Post-QoS Data Monitoring feature is not supported on Q100 ASIC-based line cards.

Flow Exporter

- Netflow exporter packet does not support TCP.

- A source interface or source address must be configured to enable the exporter. If you do not configure a source interface, the exporter remains in a disabled state. If both a source interface and a source address are configured, the source address takes precedence.

Flow Monitor

- A valid record type such as IPv4, IPv6, or MPLS must be configured for every flow monitor map

Interfaces

- NetFlow is not supported on Bridge Virtual Interface (BVI).
- The data and flow records for GRE transit traffic do not have the output interface, source, and destination prefix lengths fields set.
- We do not recommend using the management interface to export the NetFlow packets.
- The output interface field is not updated in data and flow records when the traffic is routed through ACL-based forwarding (ABF).

IPFIX 315

- If IPFIX 315 is enabled on a line card, then all the ports on that line card should have IPFIX315 configured.
- Enable the IPFIX 315 configuration on a sub interface explicitly if the traffic for that sub interface is required to be exported.
- The incoming and outgoing interface information will point to sub-interface if routed via sub-interface.
- For IPFIX 315, the outgoing interface information may not be correct incase of packets that are multicasted or broadcasted on multiple ports.
- On a dual RP centralized system, IPFIX 315 or sFlow has to be enabled on both RP's respectively. Otherwise, the system displays an error message during configuration.

BGP Attributes - MPLS Record Types on Cisco 8010 Series Routers

- On edge nodes, only ingress direction is supported.
- The MPLS record types, such as mpls ipv4-fields, mpls ipv6-fields, and mpls ipv4-ipv6-fields, are only supported.
- At the edge node on MPLS or SR-MPLS core with Penultimate Hop Popping (PHP) or Ultimate Hop Popping (UHP) disabled, the packets that are sampled should contain the MPLS label to fetch the BGP attributes.

Comparative Overview of NetFlow Version 9 and Version 10 (IPFIX)

Multiple versions of the NetFlow protocol exist. This section provides a comprehensive overview of the distinct versions within the NetFlow monitoring protocol, including NetFlow v9 and NetFlow v10 (IPFIX). It highlights the variations between these protocols.

NetFlow Version 9

NetFlow Version 9 is a template-based approach that provides flexibility in the record format. It enables enhancements to NetFlow services without concurrently altering the basic flow-record format.

NetFlow Options Template

The NetFlow Options Template serves as a distinctive template record designed to communicate the format of data associated with the NetFlow operation. Instead of sharing details about IP flows, these options serve the purpose of providing metadata pertaining to the NetFlow process itself. There are distinct options templates: the sampler options template and the interface options template. The NetFlow process exports these two tables. Furthermore, the NetFlow process also exports the VRF (Virtual Routing and Forwarding) table.

Sampler Table

The Sampler Table and Interface Option Templates play a significant role in organizing information.

The Sampler Options Template consists of a sampler table, while the Interface Option Templates consists of an interface table. Enabling these options for the sampler and interface tables simplifies the process for the collector to determine data flow information.

The sampler table offers insights into active samplers. Its primary purpose is to aid the collector in estimating the sampling rate for individual data flows. The sampler table provides the following information for each sampler:

Element ID	Field Name	Value
48	SamplerID	This ID is assigned to the sampler. It is used by the collector to retrieve information about the sampler for a data flow record.
49	SamplerMode	This field indicates the mode in which the sampling has been performed.
50	SamplerRandomInterval	This field indicates the rate at which the sampling is performed.
84	SamplerName	This field indicates the name of the sampler.

Interface Table

The interface table, contains data about interfaces that are monitored for data flow. With this data, the collector derives the interface names linked to the data flow. The interface table contains the following information:

Field Name	Value
ingressInterface	This field indicates the SNMP index assigned to the interface. By matching this value to the Ingress interface in the data flow record, the collector is able to retrieve the name of the interface.
interfaceDescription	This field indicates the name of the interface.

VRF Table

The VRF table consists mapping of VRF IDs to the VRF names. Using this information, the collector determines the name of the required VRF.

The VRF table is exported at intervals specified by the optional **timeout** keyword that can be configured manually. The default value is 1800 seconds.

The VRF table consists of the following information:

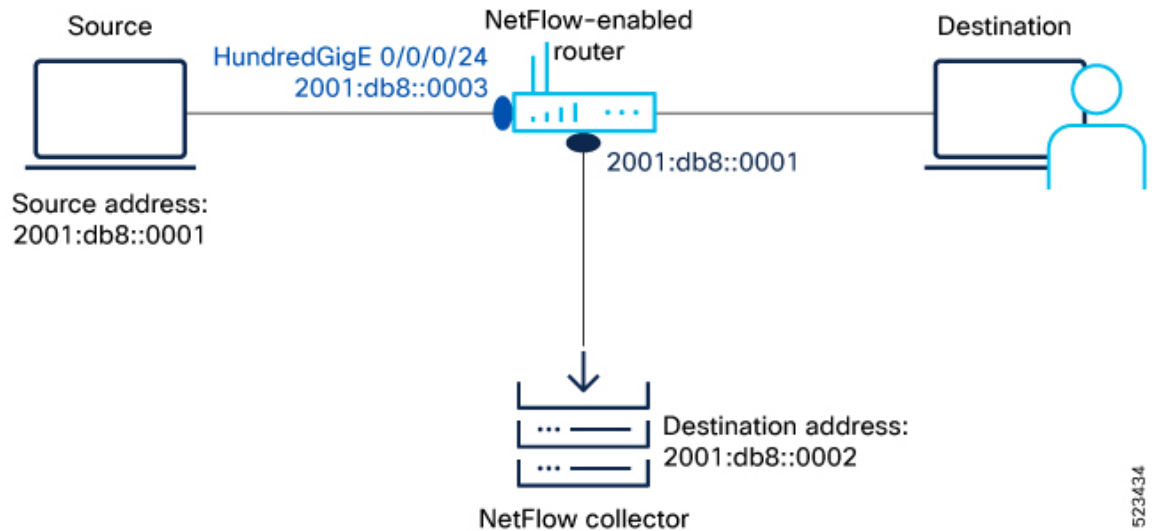
Field Name	Value
ingressVRFID	The identifier of the VRF with the name in the VRF-Name field.
VRF-Name	The VRF name has the VRFID value ingressVRFID. The value "default" indicates that the interface is not assigned explicitly to a VRF.

The data records contain ingressVRFID as an extra field in each record. The values of these fields are used to lookup the VRF Table to find the VRF names. A value of 0 in these fields indicates that the VRF is unknown.

Configure NetFlow Version 9

Let's consider the following topology to configure NetFlow.

Figure 2: NetFlow Version 9 Configuration



To monitor traffic, you must configure one or more [Flow Exporter](#), on page 6 and associate it to a Flow Monitor [Flow Monitor](#), on page 7 and enable NetFlow on the interface either in egress or ingress direction. Optionally, you can configure a [Flow Sampler](#), on page 7 to set the sampling rate for flow samples.

Before you begin

First, let's gather the required details to enable NetFlow on a router:

- The IP address of the source is : 2001:db8::0003
- The IP address of the NetFlow Collector (Destination address): 2001:db8::0002
- Interface of the router where you want to enable Netflow: HundredGigE 0/0/0/24
- The NetFlow version used to transport the data to the collector: version 9

Procedure

Step 1

Configure a Flow Exporter using the `flow exporter-map` command to specify where and how the packets should be exported.

Example:

```
Router# configure
Router(config)# flow exporter-map Exp01
Router(config-fem)# source-address 2001:db8::0003
Router(config-fem)# destination 2001:db8::0002
Router(config-fem)# transport udp 1024
Router(config-fem)# version v9
Router(config-fem-ver)# options interface-table
Router(config-fem-ver)# commit
Router(config-fem-ver)# root
Router(config)# exit
```

Step 2

Create a Flow Monitor using the `flow monitor-map` command to define the type of traffic to be monitored. You can include one or more exporter maps in the monitor map. A single flow monitor map can support up to eight exporters.

The record type specifies the type of packets that are sampled as the packets pass through the router. MPLS, IPv4, and IPv6 packet sampling is supported.

Example:

Here are the examples to record IP packets, MPLS packets, BGP packets, SRv6, and GTP-U packets.

-

```
Router#configure
Router(config)# flow monitor-map fmm-ipv6
Router(config-fmm)# record ipv6
Router(config-fmm)# cache entries 500000
Router(config-fmm)# cache timeout active 60
Router(config-fmm)# cache timeout inactive 20
Router(config-fmm)# exporter Expol
Router(config-fmm)# commit
Router(config-fmm)# root
Router(config)#exit
```

- **MPLS Packet Monitoring:** In this example, you create a flow monitor map to record the MPLS packets.

```
Router(config)#flow monitor-map fmm-mpls-ipv6
Router(config-fmm)#record mpls ipv6-fields labels 3
Router(config-fmm)#exporter Expol
Router(config-fmm)#cache entries 2000000
Router(config-fmm)#exit
```

- **BGP Packet Monitoring:** In this example, you create a flow monitor map to record the BGP packets with the permanent cache.

```
Router(config)#router bgp 50
Router(config-bgp)#address-family ipv6 unicast
Router(config-bgp-af)#bgp attribute-download
Router(config-bgp-af)#root
Router(config)#flow monitor-map fmm-bgp
Router(config-fmm)#record ipv6 peer-as
Router(config-fmm)#exporter Expol
Router(config-fmm)#cache entries 2000000
Router(config-fmm)#exit
```

- **SRv6 Packet Monitoring:** Starting from Cisco IOS-XR release 7.10.1, you can create a flow monitor map to record the SRv6 packets.

```
Router#configure
Router(config-fem)# flow monitor-map MON
Router(config-fmm)# record ipv6 srv6
Router(config-fmm)# exporter EXPRouter(config-fmm)# cache timeout inactive 5
Router(config-fmm)# !
Router(config-fmm)# sampler-map SAMP
Router(config-fmm)# random 1 out-of 1000
Router(config-fmm)# !
Router(config-fmm)# interface GigabitEthernet0/1/0/0
Router(config-fmm)# ipv6 address 2002:1::1/64
Router(config-fmm)# flow ipv6 monitor M1 sampler SAMP ingress
```

- **GTP-U Packet Monitoring:** Starting from Cisco IOS-XR release 24.2.1, you can create a flow monitor map to record the GTP-U packets.

```

Router(config)#flow monitor-map ipv6
Router(config-fmm)#record ipv6 gtp
Router(config-fmm)#exporter Exp01
Router(config-fmm)#option bgpattr
Router(config-fmm)#cache timeout active 30

Router(config-fmm)#cache timeout inactive 5

Router(config-fmm)#exit

```

Step 3 Configure a Flow Sampler using the [sampler-map](#) command to define the rate at which the packet sampling should be performed at the interface where NetFlow is enabled. Use the same sampler map configuration on the sub-interfaces and physical interfaces under a port.

Example:

```

Router(config)# configure
Router(config)# sampler-map fsm1
Router(config-sm)# random 1 out-of 262144
Router(config)# exit
Router(config)#commit
Router(config)#exit
Router#

```

Step 4 Apply a Flow Monitor Map and a Flow Sampler to a physical interface using the [flow](#) command to enable NetFlow on the router. You can choose to enable IPv4, IPv6, MPLS-aware NetFlow on the interface. Enable NetFlow in the ingress direction to monitor the incoming packets.

Note

Consider these points before applying the sampler map:

- Remove any existing Netflow or sFlow configurations before applying a new Flow sampler on an interface using the no form of the command.
- Use the same sampler map configuration on the sub-interfaces and physical interfaces under a port.

Example:

```

Router#configure
Router(config)#interface HundredGigE 0/0/0/24
Router(config-if)#flow ipv6 monitor fmm-ipv6 sampler fsm1 ingress
Router(config-if)#commit
Router(config-if)#root
Router(config)#exit

```

- **Netflow on PWHE Interface** - Apply a Monitor Map and a Sampler Map to a physical interface using the flow command to enable NetFlow on the ingress and egress of the Pseudowire Headend interface router.

```

Router(config)# interface PE1313
Router(config-if)# flow ipv6 monitor IPv6-MONITOR-MAP sampler
FNF-SAMPLER-MAP ingress
Router(config-if)# flow ipv6 monitor IPv6-MONITOR-MAP sampler
FNF-SAMPLER-MAP egress

```

Step 5 View the running configuration to verify the configuration that you have configured.

Example:

```

Router# show run

```

```

flow exporter-map Expol
version v9
  options interface-table
!
transport udp 1024
source-address 2001:db8::3
destination 2001:db8::2
!
flow monitor-map fmm-ipv6
record ipv6
exporter Expol
cache entries 500000
cache timeout active 60
cache timeout inactive 20
!
sampler-map fsm1
random 1 out-of 262144
!

interface HundredGigE0/0/0/24
shutdown
flow ipv6 monitor fmm-ipv6 sampler fsm1 ingress
!
end

```

Step 6

You can verify the the above configurations using the following steps:

- a) Verify the Flow Exporter configuration using the [show flow exporter-map](#) command.

Example:

```

Router#show flow exporter-map Expol
Flow Exporter Map : Expol
-----
Id                : 1
Packet-Length     : 1468
DestinationIpAddr : 2001:db8::2
VRFName           : default
SourceIfName      :
SourceIpAddr      : 2001:db8::3
DSCP              : 0
TransportProtocol : UDP
TransportDestPort : 1024
Do Not Fragment   : Not Enabled

Export Version: 9
  Common Template Timeout : 1800 seconds
  Options Template Timeout : 1800 seconds
  Data Template Timeout   : 1800 seconds
  Interface-Table Export Timeout : 1800 seconds
  Sampler-Table Export Timeout : 0 seconds
  VRF-Table Export Timeout  : 0 seconds

```

- b) Verify the Flow Monitor configuration using the [show flow monitor-map](#) command.

Example:

```

Router#show flow monitor-map fmm-ipv6
Flow Monitor Map : fmm-ipv6
-----
Id:                1
RecordMapName:     ipv6

```

```

ExportMapName:      Expo1
CacheAgingMode:     Normal
CacheMaxEntries:    500000
CacheActiveTout:    60 seconds
CacheInactiveTout:  20 seconds
CacheUpdateTout:    N/A
CacheRateLimit:     2000
HwCacheExists:      False
HwCacheInactTout:   50

```

- c) Verify the sampler map configuration using the [show sampler-map](#) command.

Example:

```

Router#show sampler-map fsm1

Sampler Map : fsm1
-----
Id:          1
Mode:        Random (1 out of 262144 Pkts)
Router#

```

What to do next

You can now analyse the exported data using a NetFlowAnalyser.

Verify NetFlow Version 9

Procedure

Verify the flows captured using the [show flow monitor name cache](#) command.

- **Cache Summary**

In the following example, you can verify the amount of flows added and exported.

```

Router#show flow monitor fmm-ipv6 cache summary location 0/0/CPU0
Cache summary for Flow Monitor monitor1:
Cache size: 1000000
Current entries: 295
Flows added: 184409
Flows not added: 0
Ager Polls: 9824
- Active timeout 183855
- Inactive timeout 259
- Immediate 0
- TCP FIN flag 0
- Emergency aged 0
- Counter wrap aged 0
- Total 184114
Periodic export:
- Counter wrap 0
- TCP FIN flag 0
Flows exported 184114

```

- **Cache Record for SRv6 L2 services**

This example shows the complete recorded data for SRv6 L2 services

```
Router#show flow monitor fmm-ipv6 cache record location 0/0/CPU0
===== Record number: 1 =====
IPv6SrcAddr : 2::2
IPv6DstAddr : bbbb:bc00:88:e000::
BGPDstOrigAS : 0
BGPSrcOrigAS : 0
BGPNextHopV6 : fe80::232:17ff:fe7e:1ce1
IPv6TC : 0
IPv6FlowLabel : 50686
IPv6OptHdrs : 0x0
IPv6Prot : 143
L4SrcPort : 0
L4DestPort : 0
L4TCPFlags : 0
IPv6DstPrfxLen : 48
IPv6SrcPrfxLen : 128
InputInterface : Hu0/0/0/10
OutputInterface : BE11.1
ForwardStatus : Fwd
FirstSwitched : 01 18:51:25:797
LastSwitched : 01 18:51:25:797
ByteCount : 61004304
PacketCount : 113814
Dir : Ing
SamplerID : 1
InputVRFID : default
OutputVRFID : default
InnerIPv4SrcAddr : 0.0.0.0
InnerIPv4DstAddr : 0.0.0.0
InnerIPv6SrcAddr : ::
InnerIPv6DstAddr : ::
InnerL4SrcPort : 0
InnerL4DestPort : 0
SrcMacAddr : 00:0c:29:0e:d8:32
DstMacAddr : 00:0c:29:0e:d8:3c
EthType : 2048
Dot1qPriority : 0
Dot1qVlanId : 2001
RecordType : SRv6 L2 Service Record
SRHFlags : 0x0
SRHTags : 0x0
SRHSegmentsLeft : 0
SRHNumSegments : 0
```

• Cache Record for QoS services

In the following example, you can verify the QoS value from the IPv6TC field and post QoS type of service (TOS)/DSCP from the IPv6PostTC field.

```
Router#show flow monitor fmm-ipv6 cache record location 0/0/CPU0
===== Record number: 1 =====
IPv6SrcAddr : 2001:db8::3
IPv6DstAddr : 2001:db8::2
BGPDstOrigAS : 0
BGPSrcOrigAS : 0
BGPNextHopV6 : 1000:1::2
IPv6TC : 0
IPv6PostTC : 48
IPv6FlowLabel : 0
IPv6OptHdrs : 0x10
IPv6Prot : 59
```

```

L4SrcPort      : 0
L4DestPort     : 0
L4TCPFlags     : 0
IPv6DstPrfxLen : 128
IPv6SrcPrfxLen : 128
InputInterface : FH0/0/0/0
OutputInterface : FH0/0/0/12
ForwardStatus  : Fwd
FirstSwitched  : 00 00:59:02:416
LastSwitched   : 00 00:59:02:416
ByteCount      : 46
PacketCount    : 1
Dir            : Ing
SamplerID      : 1
InputVRFID     : default
OutputVRFID    : default

```

• Cache Record for GTP-U services

In the following example, you can verify the GTP tunnel ID, QoS flow identifier, and GTP session details from the GTPTeid, GTPQFI and GTPSESSDIR field.

```
Router#show flow monitor fmm-ipv6 cache format record location 0/0/CPU0
```

```

===== Record number: 1 =====
RecordType      : GTP Tunneled Record
IPv4SrcAddr     : 0.0.0.0
IPv4DstAddr     : 0.0.0.0
IPv6SrcAddr     : 2001:db8:1::1
IPv6DstAddr     : 2001:db8:2::2
L4SrcPort      : 0
L4DestPort     : 0
IPv4Prot       : icmpv6
IPv4TOS        : 0
InputInterface : Gi0/2/0/0
OutputInterface : 0
L4TCPFlags     : 0
ForwardStatus  : Fwd
FirstSwitched  : 00 00:08:59:286
LastSwitched   : 00 00:08:59:286
ByteCount      : 1296
PacketCount    : 1
Dir            : Ing
GTPTeid       : 11
GTPQFI        : 0
GTPSESSDIR    : 0
IPv6TC         : 0
IPv6FlowLabel   : 690680
MinimumTTL      : 64
MaximumTTL      : 64
IPFragFlags     : 0
IPFragOffset    : 181
IPIdentification : 0
IPv6Ident       : 1546089621
L4SequenceNum   : 0
L4Checksum      : 0
MinPktLen       : 100
MaxPktLen       : 100
ICMPBytes       : 0x8000cf945edf0002
OuterIPv4SrcAddr : 100.100.100.1
OuterIPv4DstAddr : 200.200.200.2
OuterIPv6SrcAddr : ::
OuterIPv6DstAddr : ::
BGPNextHopV4    : 0.0.0.0
BGPNextHopV6    : ::

```



```

BGPSrcOrigAS      : 0
BGPDstOrigAS      : 0
IPV4NextHop        : 0.0.0.0
IPV6NextHop        : ::
SrcMacAddr         : 00:00:3f:11:50:20
DstMacAddr         : 45:00:00:62:00:00
EthType            : 2048
Dot1qPriority       : 0
Dot1qVlanId        : 0
CustVlanId         : 0
InputVRFID         : default
OutputVRFID        : default
===== Record number: 2 =====
RecordType         : GTP Tunneled Record
IPV4SrcAddr        : 192.168.12.2
IPV4DstAddr        : 192.168.12.1
IPv6SrcAddr        : ::
IPv6DstAddr        : ::
L4SrcPort          : 0
L4DestPort         : 0
IPV4Prot           : icmp
IPV4TOS            : 0
InputInterface     : Gi0/2/0/0
OutputInterface    : 0
L4TCPFlags         : 0
ForwardStatus      : Fwd
FirstSwitched      : 00 00:08:54:244
LastSwitched       : 00 00:08:54:244
ByteCount          : 64
PacketCount        : 1
Dir                : Ing
GTPTeid          : 11
GTPQFI           : 0
GTPSESSDIR       : 0
IPv6TC             : 0
IPv6FlowLabel      : 0
MinimumTTL         : 255
MaximumTTL         : 255
IPFragFlags        : 0
IPFragOffset       : 97
IPIdentification   : 4
IPV6Ident          : 0
L4SequenceNum      : 0
L4Checksum         : 0
MinPktLen          : 100
MaxPktLen          : 100
ICMPBytes          : 0xabcdabcdabcdabcd
OuterIPV4SrcAddr   : 100.100.100.1
OuterIPV4DstAddr   : 200.200.200.2
OuterIPv6SrcAddr   : ::
OuterIPv6DstAddr   : ::
BGPNextHopV4       : 0.0.0.0
BGPNextHopV6       : ::
BGPSrcOrigAS       : 0
BGPDstOrigAS       : 0
IPV4NextHop        : 0.0.0.0
IPV6NextHop        : ::
SrcMacAddr         : 00:00:3f:11:50:20
DstMacAddr         : 45:00:00:62:00:00
EthType            : 2048
Dot1qPriority       : 0
Dot1qVlanId        : 0
CustVlanId         : 0

```

```

InputVRFID      : default
OutputVRFID     : default

```

What to do next

You can now analyse the exported data using a NetFlowAnalyser.

Modify NetFlow Configuration

You can modify only the following flow attributes that is already applied to an interface for a monitor map, exporter map, or a sampler map.

Note that when you modify the flow attributes, the cache counters are cleared and results in resetting of the counters. As a result there could be flow accounting mismatch.

Table 6: Flow Entities and Flow Attributes that can be altered

Flow Entity	Flow Attribute	Command
Monitor map	cache timeout	cache timeout
	• active • inactive • update • rate-limit	
	exporter	exporter
	cache entries	cache entries
	cache permanent	cache permanent
Exporter Map	options outphysint bgpattr filtered outbundlemember	options
	source <source interface>	source
	destination <destinaiton address>	destination
	dscp <dscp_value>	dscp
Sampler Map	version v9 ipfix	version ipfix
	sampling interval	sampling interval

IPFIX (NetFlow version 10)

Table 7: Feature History Table

Feature Name	Release Information	Feature Description
IPFIX flow record enhancements for L2 and L3 traffic	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I
IPFIX flow record enhancements for L2 and L3 traffic	Release 24.4.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). The IPFIX flow record enhancements for L2 and L3 traffic introduced in Cisco IOS XR Release 7.2.12 to ensure scalability and adaptability to diverse network environments are now supported on the following hardware. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH
IPFIX flow record enhancements for L2 and L3 traffic	Release 7.2.12	This release introduces: • Support for flow-based IPFIX protocol version 10(v10), for L2 interfaces. Only L3 interfaces were supported in previous releases. • A new record-type, MPLS-IPv4, to capture BGP next-hop information.

Internet Protocol Flow Information Export (IPFIX) has been standardized by the Internet Engineering Task Force (IETF) as an export protocol for transmitting NetFlow packets. Building upon NetFlow version 9, IPFIX introduces efficient flow data formatting through templates, ensuring scalability and adaptability to diverse network environments. Utilizing UDP as the transport protocol, IPFIX facilitates the seamless transfer of NetFlow information from exporters to collectors. With native support for IPv6 flow records, the inclusion of optional data fields, and the ability to send data to multiple collectors, IPFIX proves to be a versatile and powerful solution for network administrators, enabling comprehensive traffic analysis, monitoring, and enhanced visibility into network behavior.

IPFIX 315

The Internet Engineering Task Force (IETF) has standardized Internet Protocol Flow Information Export (IPFIX) as an export protocol for sending IP flow information. Router supports the IPFIX 315 format for exporting flow information. The IPFIX 315 format enables the transmission of 'n' octets of frame information starting from the Ethernet header up to the transport header of the traffic flow over the network. IPFIX 315 supports the sending of variable-sized packet records with variable payload information, such as IPv4, IPv6, MPLS, and nested packets like OuterIP-GRE-InnerIP, and more. The process involves sampling and exporting the traffic flow information. Also, along with the Ethernet frame information, the IPFIX 315 format exports the information of the incoming and outgoing interfaces of the sampled packet.

The information of the packets flowing through a device is used for a variety of purposes, including network monitoring, capacity planning, traffic management, and more.

When exporting packets, a special cache-type called Immediate Aging is used. Immediate Aging ensures that the flows are exported as soon as they are added to the cache.

Sampling and Exporting Information

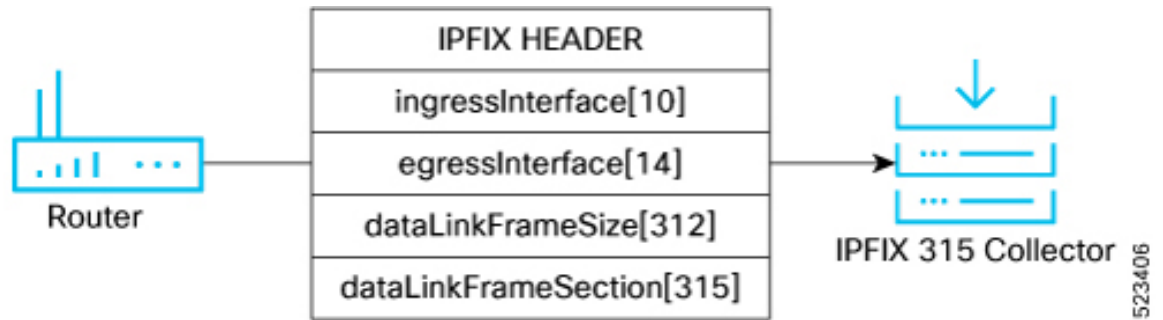
To sample the traffic flow information, configure a sampler-map that specifies the rate at which packets (one out of every 'n' packets) are sampled. Not all packets flowing through a device are exported; only the packets selected based on the sampling rate are exported.

The size of the exported packet depends on the sampled packet size and the location of the L4 header. The exported packet size is determined as follows:

- If the sampled packet size is more than 160 bytes and the L4 header is not obtained within the first 160 bytes, the exported packet size is 160 bytes.
- If the L4 header is within the first 160 bytes, the exported packet size is equal to the length of the sampled packet until the L4 header.
- If the packet size is less than 160 bytes and the L4 header isn't within the first 160 bytes, the exported packet size is equal to the length of the packet.

This figure *IPFIX 315 Export Packet Format* shows exported packet information.

Figure 3: IPFIX 315 Export Packet Format



Monitoring Post-QoS data in NetFlow and IPFIX

Table 8: Feature History Table

Feature Name	Release Information	Description
Monitoring post-QoS data in NetFlow and IPFIX	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I
Monitoring post-QoS data in NetFlow and IPFIX	Release 24.4.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). This feature which empowers you to verify that QoS policies are applied as required, obtain a comprehensive view of your network's performance, and gain detailed insights into how QoS policies influence traffic is now supported on the following hardware. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH

Feature Name	Release Information	Description
Monitoring post-QoS data in NetFlow and IPFIX	Release 24.1.1	You now have the capability to obtain a comprehensive view of your network's performance, gaining detailed insights into how QoS policies influence traffic. This empowers you to verify that QoS policies are applied as expected and assess the effectiveness of QoS mechanisms in prioritizing critical traffic. Access to this information is made possible by enabling the export of post-QoS information in both Netflow and IPFIX, supported by the incorporation of Post-QoS Information Elements.

You can now monitor information on QoS policies through the post-QoS information element field export for NetFlow and IPFIX. This improvement focuses on the analysis of packet-level information by incorporating the export of Post-QoS details, specifically the Differentiated Services Code Point (DSCP) in IPv4 and Traffic Class in IPv6. This enhancement facilitates in-depth monitoring of sampled packets, emphasizing their post-processing QoS characteristics.

In NetFlow v9 or IPFIX packets, the post-QoS field utilizes the information element postIpClassOfService (IE55) for collecting post-QoS data. Moreover, this feature brings added visibility to both pre- and post-QoS Type of Service (TOS)/DSCP values within the show command, providing you with the data to assess network performance comprehensively.



Note The ingress QoS (i.e., pre-QoS) parameters are already exported in prior IOS-XR software releases.

- **ipClassOfService** - For IPv4, it represents the value of the Type of Service (TOS) field in the IPv4 packet header. For IPv6 packets, it signifies the value of the Traffic Class field in the IPv6 packet header. This information element is available prior to IOS-XR software release 24.1.1.
- **postIpClassOfService** - The definition of this information element is identical to the above mentioned definition of 'ipClassOfService,' except that it reports a potentially modified value caused by a router function after the packet has passed the observation point. This information element is introduced in IOS-XR software release 24.1.1.

NetFlow v9 and NetFlow v10 (IPFIX)

This section helps you understand the NetFlow v9 and NetFlow v10 (IPFIX) based on the following factors:

Table 9: NetFlow v9 and NetFlow v10 (IPFIX)

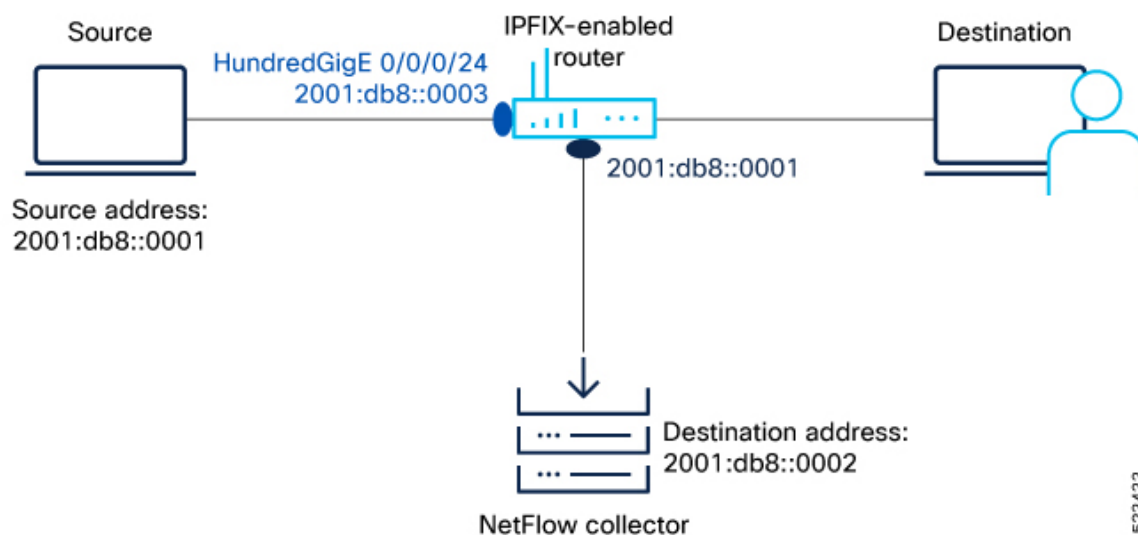
Factor	NetFlow v9	NetFlow v10 (IPFIX)
Transport	Typically uses UDP	Supports both UDP and TCP transport protocols

Factor	NetFlow v9	NetFlow v10 (IPFIX)
Compatibility	Compatible with older NetFlow versions	Backward-compatible with NetFlow v9
Flexibility	Fixed set of predefined fields	More flexible with variable-length information elements and custom-defined attribute
Information Elements	Limited set of predefined fields	Extensive list of predefined elements

Configure IPFIX

Let's consider the following topology to configure IPFIX:

Figure 4: NetFlow IPFIX Configuration



To monitor traffic, you must configure one or more [Flow Exporter, on page 6](#) and associate it to a [Flow Monitor, on page 7](#) and enable IPFIX on the interface either in egress or ingress direction. Optionally, you can configure a [Flow Sampler, on page 7](#) to set the sampling rate for flow samples.

Procedure

Step 1 First, let's gather the required details to enable IPFIX on a router:

- The IP address of the source : 2001:db8::0001
- The IP address of the IPFIX Collector (Destination address): 2001:db8::0002
- Interface of the router where we will enable IPFIX: HundredGigE 0/0/0/24
- NetFlow version used to transport the data to the collector: IPFIX

Step 2 Configure a Flow Exporter using the `flow exporter-map` command to specify where and how the packets should be exported.

```
Router(config)# flow exporter-map fem_ipfix
Router(config-fem)# destination 2001:db8::0002
Router(config-fem)# source Loopback 0
Router(config-fem)# transport udp 9001
Router(config-fem)# version ipfix
Router(config-fem-ipfix)# template data timeout 600
Router(config-fem-ipfix)# options interface-table
Router(config-fem-ipfix)# exit
```

Verify the Flow Exporter configuration using the `show flow exporter-map` command.

```
Router#show flow exporter-map fem_ipfix
Flow Exporter Map : fem_ipfix
-----
Id                : 1
Packet-Length     : 1468
DestinationIpAddr : 2001:db8::2
VRFName           : default
SourceIfName      :
SourceIpAddr      : 2001:db8::3
DSCP              : 0
TransportProtocol : UDP
TransportDestPort : 1024
Do Not Fragment   : Not Enabled

Export Version: IPFIX
Common Template Timeout : 1800 seconds
Options Template Timeout : 1800 seconds
Data Template Timeout : 1800 seconds
Interface-Table Export Timeout : 1800 seconds
Sampler-Table Export Timeout : 0 seconds
VRF-Table Export Timeout : 0 seconds
```

Step 3 Create a Flow Monitor using the `flow monitor-map` command to define the type of traffic to be monitored. You can include one or more exporter maps in the monitor map. A single flow monitor map can support up to eight exporters. The record type specifies the type of packets that are sampled as the packets pass through the router. MPLS, IPv4, and IPv6 packet sampling is supported.

```
Router(config)# flow monitor-map fmm1
Router(config-fmm)# record ipv6
Router(config-fmm)# option filtered
Router(config-fmm)# exporter fem_ipfix
Router(config-fmm)# cache entries 65535
Router(config-fmm)# cache timeout active 1800
Router(config-fmm)# cache timeout inactive 15
Router(config-fmm)# exit
```

Verify the Flow Monitor configuration using the `show flow monitor-map` command.

```
Router#show flow monitor-map fmm1
Flow Monitor Map : fmm1
-----
Id:                1
RecordMapName:     ipv6
ExportMapName:     Expo1
CacheAgingMode:    Normal
CacheMaxEntries:   500000
```



```
CacheActiveTout: 60 seconds
CacheInactiveTout: 20 seconds
CacheUpdateTout: N/A
CacheRateLimit: 2000
HwCacheExists: False
HwCacheInactTout: 50
```

- Step 4** Configure a Flow Sampler using the `sampler-map sampler-map` command. Use the same sampler map configuration on the sub-interfaces and physical interfaces under a port.

```
Router(config)# configure
Router(config)# sampler-map fsm1
Router(config-sm)# random 1 out-of 4000
Router(config)# exit
Router(config)# commit
Router(config)# exit
Router#
```

Verify the sampler map configuration using the `show sampler-map show sampler-map` command.

```
Router#show sampler-map fsm1

Sampler Map : fsm1
-----
Id:      1
Mode:    Random (1 out of 4000 Pkts)
Router#
```

- Step 5** View the running configuration to verify the configuration that you have configured.

```
Router#show run
Thu Nov 9 06:40:07.296 UTC
Building configuration...

flow exporter-map fem_ipfix
version ipfix
options interface-table
template data timeout 600
!
transport udp 9001
source Loopback0
destination 2001:db8::2
!
flow monitor-map fmm1
record ipv6
option filtered
exporter fem_ipfix
cache entries 65535
cache timeout active 1800
cache timeout inactive 1
!
sampler-map fsm1
random 1 out-of 4000
!
interface FourHundredGigE0/0/0/1
shutdown
!
interface HundredGigE0/0/0/24
shutdown
flow ipv4 monitor fmm1 sampler fsm1 ingress
flow ipv6 monitor fmm-ipv6 sampler fsm1 ingress
!
end
```

- Step 6** Apply a Monitor Map and a Sampler Map to a physical interface using the [flow](#) command to enable IPFIX on the router.

```
Router(config)#interface HundredGigE 0/0/0/24
Router(config-if)#flow ipv4 monitor fmm1 sampler fsm1 ingress
Router(config-if)#exit
```

Configure IPFIX 315

This section provides you instructions to enable IPFIX 315 on Cisco IOS XR Software.

Procedure

- Step 1** Enable IPFIX 315 for flow monitoring.

```
Router(config)# hw-module profile netflow ipfix315-enable
```

- Step 2** Configure an exporter map with IPFIX as the exporter version using the [flow exporter-map](#) command in global configuration mode to specify where and how the packets should be exported.

```
Router(config)# flow exporter-map ipfix_exp
Router(config-fem)# version ipfix
Router(config-fem-ipfix)# template data timeout 10
Router(config-fem)# dscp 63
Router(config-fem)# transport udp 12000
Router(config-fem)# source Loopback 0
Router(config-fem)# destination 100.10.1.159
Router(config-fem)# exit
```

- Step 3** Create a flow monitor using the [flow monitor-map](#) command in global configuration mode to define the type of traffic to be monitored. You can include one or more exporter maps in the monitor map.

```
Router(config)# flow monitor-map ipfix_mon
Router(config-fmm)# record datalinksectiondump
Router(config-fmm)# exporter ipfix_exp
Router(config-fmm)# cache immediate
Router(config-fmm)# exit
```

- Step 4** Configure a sampler map using the [sampler-map](#) command to define the rate at which the packet sampling should be performed at the interface where IPFIX is enabled.

```
Router# sampler-map ipfix_sm
Router(config-sm)# random 1 out-of 32000
Router(config)# exit
```

- Step 5** Apply a monitor map and a Sampler Map to a physical interface using the [flow](#) command to enable IPFIX on the router.

```
Router(config)#interface TenGigE0/0/0/5
Router(config-if)#ipv4 address 192.1.108.2 255.255.255.0
Router(config-if)#ipv6 address 1:108::2/64
Router(config-if)#flow datalinkframesection monitor ipfix_mon sampler ipfix_sm ingress
Router(config-if)#encapsulation dot1q 139
```

- Step 6** Verify the sampled and exported flow statistics using the [show flow platform producer statistics location](#) command.

In this show output, you can see that the system has actively received and monitored a total of 630,478 IPFIX 315 packets.

```
Router#show flow platform producer statistics location 0/0/0/5
Netflow Platform Producer Counters:
IPv4 Ingress Packets: 0
IPv4 Egress Packets: 0
IPv6 Ingress Packets: 0
IPv6 Egress Packets: 0
MPLS Ingress Packets: 0
MPLS Egress Packets: 0
IPFIX315 Ingress Packets: 630478
IPFIX315 Egress Packets: 0
Drops (no space): 0
Drops (other): 0
Unknown Ingress Packets: 0
Unknown Egress Packets: 0
Worker waiting: 2443
```

Step 7 Verify the flow monitor stats statistics using the [show flow monitor cache location](#) command.

This example shows that there were 50399 flows added to the cache and exported.

```
Router#show flow platform producer statistics location 0/0/0/5
Cache summary for Flow Monitor ipfix_mon:
Cache size: 65535
Current entries: 0
Flows added: 50399
Flows not added: 0
Ager Polls: 2784
- Active timeout 0
- Inactive timeout 0
- Immediate 50399
- TCP FIN flag 0
- Emergency aged 0
- Counter wrap aged 0
- Total 50399
Periodic export:
- Counter wrap 0
- TCP FIN flag 0
Flows exported 50399
Matching entries: 0
```

Collect additional BGP information elements for MPLS IPv4 and IPV6 using IPFIX

Table 10: Feature History Table

Feature Name	Release Information	Description
Collect additional BGP information elements for MPLS IPv4 and IPV6 using IPFIX	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I

Feature Name	Release Information	Description
Collect additional BGP information elements for MPLS IPv4 and IPv6 using IPFIX	Release 24.4.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). This feature helps you monitor and optimize your network more effectively with IPFIX, which enhances the collection of BGP Information Elements (IEs) in IPFIX records. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH

This feature is specifically designed to improve congestion mitigation in core-edge link scenarios, this update introduces support for gathering eight additional BGP fields in IPFIX MPLS IPv4/IPv6 records. With this feature, you can efficiently monitor and optimize your network.

Additionally, two new Information Elements, namely Minimum Time-to-Live (TTL) and Maximum TTL, are recorded. These elements provide information about the minimum Time to Live for a flow and the maximum Time to Live for a flow.

Table 11: Information Elements

IE Field	IE Number
BgpSourceAsNumber	16
BgpDestinationAsNumber	17
BgpNextHopIPv4Address	18
BgpNextHopIPv6Address	63
DestinationIPv4PrefixLength	13
DestinationIPv6PrefixLength	30
IpNextHopIPv4Address	15
IpNextHopIPv6Address	62
Minimum TTL	52
Maximum TTL	53

IE number, or Information Element Number, is a unique identifier assigned to specific elements within network communication protocols, facilitating standardized interpretation and management. For more information refer [IP Flow Information Export \(IPFIX\) Entities](#).

Configuration

The following example shows how to collect MPLS traffic with both IPv6 and IPv4 fields.

Configuring Monitor map:

```
Router(config)#flow monitor-map mpls-1
Router(config-fmm)#record mpls ipv4-ipv6-fields
Router(config-fmm)#commit
Router(config-fmm)#exit
```

Configuring Sampler map:

```
Router(config)#sampler-map fsm1
Router(config-sm)#random 1 out-of 4000
Router(config-sm)#commit
Router(config-sm)#exit
```

Apply a Monitor Map and a Sampler Map to a physical interface

```
Router(config)#interface HundredGigE 0/0/0/24
Router(config-if)#flow mpls monitor mpls-1 sampler fsm1 ingress
Router(config-if)#exit
```

Verification

Verify the flow monitor stats statistics using the **show flow monitor cache location command**.

```
Router#show flow monitor mpls-1 cache summary location 0/0/CPU0===== Record number: 1
=====
===== Record number: 1 =====
LabelType       : Unknown
Prefix/Length   : 20.1.1.0/24
Label1-EXP-S    : 16001-0-1
Label2-EXP-S    : -
Label3-EXP-S    : -
Label4-EXP-S    : -
Label5-EXP-S    : -
Label6-EXP-S    : -
InputInterface  : FH0/0/0/1
OutputInterface : FH0/0/0/0
ForwardStatus   : Fwd
FirstSwitched   : 00 08:28:52:189
LastSwitched    : 00 08:28:57:649
ByteCount       : 2352
PacketCount     : 56
Dir             : Ing
SamplerID       : 1
IPv4SrcAddr     : 30.1.1.1
IPv4DstAddr     : 20.1.1.1
IPv4TOS         : 0
IPv4Prot        : udp
L4SrcPort       : 2025
L4DestPort      : 2500
L4TCPFlags      : 0
IPv4SrcPrfxLen  : 24
IPv4DstPrfxLen  : 24
BGPNextHopV4    : 192.168.10.10
BGPNextHopV6    : ::
BGPSrcOrigAS    : 2000
```

Collect additional BGP information elements for MPLS IPv4 and IPv6 using IPFIX

```

BGPDstOrigAS      : 1000
IPv4NextHop       : 192.168.10.10
IPv6NextHop       : ::
MinimumTTL        : 90
MaximumTTL        : 110
InputVRFID        : default
OutputVRFID       : default

```

===== Record number: 1 =====

```

LabelType         : Unknown
Prefix/Length     : ::/0
Label1-EXP-S      : 16001-0-1
Label2-EXP-S      : -
Label3-EXP-S      : -
Label4-EXP-S      : -
Label5-EXP-S      : -
Label6-EXP-S      : -
InputInterface    : FH0/0/0/1
OutputInterface   : FH0/0/0/0
ForwardStatus     : Fwd
FirstSwitched     : 00 08:27:38:692
LastSwitched      : 00 08:27:47:572
ByteCount         : 5580
PacketCount       : 90
Dir               : Ing
SamplerID         : 1
IPv6SrcAddr       : 50::1
IPv6DstAddr       : 40::1
IPv6TC            : 0
IPv6FlowLabel     : 0
IPv6OptHdrs       : 0x0
IPv6Prot          : udp
L4SrcPort         : 2025
L4DestPort        : 2500
L4TCPFlags        : 0
IPv6SrcPrfxLen    : 64
IPv6DstPrfxLen    : 64
BGPNextHopV4      : 0.0.0.0
BGPNextHopV6      : ::ffff:192.168.10.10
BGPSrcOrigAS      : 2000
BGPDstOrigAS      : 1000
IPv4NextHop       : 192.168.10.10
IPv6NextHop       : ::
MinimumTTL        : 195
MaximumTTL        : 205
InputVRFID        : default
OutputVRFID       : default

```



Note When processing the ICMP Layer 4 header, the destination port is determined based on the ICMPv6 message type, instead of being set to zero. This behavior is specific to ICMPv6 and does not apply to ICMP for IPv4.

For example, in the following output, the L4DestPort value corresponds to ICMPv6 Msg Type 3 (Time Exceeded). See <https://www.iana.org/assignments/icmpv6-parameters/icmpv6-parameters.xhtml#icmpv6-parameters-codes-4>

```
IPv6SrcAddr      : 1700::2
IPv6DstAddr      : 1800::2
BGPDstOrigAS     : 0
BGPSrcOrigAS     : 0
BGPNextHopV6     : fcbb:bb00:3::
IPv6TC           : 8
IPv6FlowLabel    : 4
IPv6OptHdrs      : 0x0
IPv6Prot         : icmpv6
MinimumTTL       : 120
MaximumTTL       : 120
L4SrcPort        : 0
L4DestPort       : 3
L4TCPFlags       : 0
IPv6DstPrfxLen   : 64
IPv6SrcPrfxLen   : 128
InputInterface   : Hu0/0/0/1
OutputInterface  : Hu0/0/0/0
ForwardStatus    : Fwd
BGPNextHopV4     : 0.0.0.0
IPv4NextHop      : 0.0.0.0
IPv6NextHop      : ::
FirstSwitched    : 04 02:36:55:363
LastSwitched     : 04 02:37:19:963
ByteCount        : 25190
PacketCount      : 229
Dir              : Ing
SamplerID        : 4
SrcMacAddr       : 00:ca:ff:ee:00:01
DstMacAddr       : 04:00:00:07:1d:04
EthType          : 34525
Dot1qPriority     : 0
Dot1qVlanId      : 0
CustVlanId       : 0
InputVRFID       : vrf_1
OutputVRFID      : default
```

BGP community and AS path information elements for IPFIX

BGP community and AS path information elements are IPFIX (NetFlow v10) data elements that

- enable tagging of network flows with BGP community and AS path values
- allow users to correlate flow records with BGP path attributes for both communities and autonomous system numbers, and
- support enhanced flow analysis, troubleshooting, and policy verification.

BGP communities and AS paths

BGP communities are mechanisms that tag routes with additional information, making it easier for network operators to manage routing policies and analyze routing decisions based on BGP attributes.

AS path represents the sequence of autonomous systems a route has traversed, providing insight into the journey of a route across multiple networks and enabling more informed routing decisions.

Table 12: Feature History Table

Feature Name	Release Information	Feature Description
BGP community and AS path information elements for IPFIX	Release 25.3.1	Introduced in this release on: Fixed Systems (8200 [ASIC: Q200, P100], 8700 [ASIC: P100, K100], 8010 [ASIC: A100]); Centralized Systems (8600 [ASIC: Q200]); Modular Systems (8800 [LC ASIC: Q100, Q200, P100]) You can now export flow records with BGP community and AS path information elements in IPFIX. This allows you to monitor and analyze traffic based on precise BGP routing metadata, including community tags and AS paths. This ensures more granular visibility and control over your routing environment.

Exported attributes information

The IPFIX (NetFlow v10) export supports two primary BGP attributes:

- `bgpDestinationCommunityList` (IE485): Exports a list of BGP community values associated with the destination of each flow.
- `bgpDestinationAsPathList` (IE512): Exports the sequence of autonomous systems (AS path) associated with the destination prefix of each flow.

Guidelines for exporting BGP and AS attributes using IPFIX

Requirements for exporting BGP and AS attributes using IPFIX

- Set the exporter map to IPFIX version to export BGP attributes.
- Exported list is limited to 32 elements. If a list contains more than 32 elements, it will be truncated, and only the first 32 elements will be exported.
- Verify BGP attribute export by checking the exported packets at the collector because show commands do not display these changes.

Caution for exporting BGP and AS attributes using IPFIX

- This export feature is specific to NetFlow v10 (IPFIX).

Configure BGP community and AS path export for IPFIX

Use this procedure when you need to correlate IPFIX traffic records with BGP routing attributes. This is required for advanced traffic analysis, troubleshooting, or policy enforcement use cases

Before you begin

Before starting this procedure, make sure you have completed the necessary prerequisites.

- Ensure you are running release 25.3.1 or later.
- Confirm that your platform supports IPFIX and the relevant flow monitor and exporter configurations.

Follow these steps to configure BGP community and AS path export for IPFIX:

Procedure

Step 1 Configure the Flow Monitor Map for IPv4 Recording and Export of BGP Community and AS Path Attributes.

```
Router# configure
Router(config)# flow monitor-map fmm-test
Router(config-fmm)# record ipv4
Router(config-fmm)# option bgpattr
Router(config-fmm)# exporter fem-test
Router(config-fmm)# exit
```

Step 2 Verify that your exporter map specifies version ipfix.

```
Router(config)# flow exporter-map fem-test
Router(config-fem)# version ipfix
Router(config-fem)# transport udp 2055
Router(config-fem)# source-address 10.0.0.2
Router(config-fem)# destination 10.0.0.1
Router(config-fem)# exit
```

Step 3 Configure the sample map.

```
Router(config)# sampler-map fsm-test
Router(config-samplermap)# random 1 out-of 8000
Router(config-samplermap)# exit
```

Step 4 Apply the flow monitor map to the required interface.

```
Router(config)# interface FourHundredGigE0/0/0/17
Router(config-if)# flow ipv4 monitor fmm-test sampler fsm-test ingress
Router(config-if)# end
```

Step 5 Verify BGP community and AS path attribute export.

```
Router# show run flow monitor-map fmm-test
Fri Aug 29 05:40:58.443 UTC
flow monitor-map fmm-test
  record ipv4
  option bgpattr
```

```
exporter fem-test
!
```

BGP community and AS path information elements are included in exported IPFIX flow records.

Revision History

Table 13: Feature History Table

Feature Name	Release Information	Feature Description
Monitor GTP-U Traffic in 5G Network	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I
Monitor GTP-U Traffic in 5G Network	Release 24.4.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100, K100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) You now get a comprehensive view of your 5G network's performance and gain detailed insights into the slice utilization, QoS policies applied, and their impact on traffic. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-36EH • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM

Feature Name	Release Information	Feature Description
Monitor GTP-U Traffic in 5G Network	Release 24.2.11	You now get a comprehensive view of your 5G network's performance and gain detailed insights into the slice utilization, QoS policies applied, and their impact on traffic. This includes verifying the QoS policies of the deployed slices, assessing the effectiveness of 5G slice mechanisms and tracking GTP-U endpoints for specific applications or services. This information is available because we've enabled the exporting of GTP-U related Information Elements. This feature introduces these changes: CLI: The gtp keyword is introduced in the record ipv4 and record ipv6 commands.
Collect Additional BGP Information Elements for MPLS IPv4 and IPV6 Using IPFIX	Release 24.1.1	You can now collect insights into how MPLS traffic is flowing through the network, assess the performance of your traffic engineering policies and make informed adjustments, pinpoint where in your MPLS network packets are being misrouted or dropped for swift troubleshooting, and also enable accurate billing for your users' customers because of insights into accurate resource usage. This is made possible because we have enabled the collection of BGP information elements for MPLS IPv4 and IPv6 traffic using IPFIX. This feature modifies the output of the show flow monitor command.

Feature Name	Release Information	Feature Description
Monitoring Post-QoS Data in NetFlow and IPFIX	Release 24.1.1	You now have the capability to obtain a comprehensive view of your network's performance, gaining detailed insights into how QoS policies influence traffic. This empowers you to verify that QoS policies are applied as expected and assess the effectiveness of QoS mechanisms in prioritizing critical traffic. Access to this information is made possible by enabling the export of post-QoS information in both Netflow and IPFIX, supported by the incorporation of Post-QoS Information Elements.
IPFIX Enablement for SRv6 and Services over SRv6 Core	Release 7.10.1	During the transition from conventional IP/MPLS networks to SRv6-based networks, the necessity for monitoring SRv6 traffic flow becomes crucial. This feature enables IPFIX to effectively monitor SRv6 IP traffic flow from network devices in the core. The feature introduces these changes: CLI: The srv6 keyword is introduced in the record ipv6 command.



CHAPTER 6

sFlow Configuration for Traffic Monitoring and Analysis

This page helps you with information about the key concepts, advantages and limitations of sFlow, and steps to configure sFlow on your router.

- [sFlow Essential Concepts and Terms, on page 63](#)
- [Flow Monitoring on Egress Interface, on page 64](#)
- [How sFlow works, on page 65](#)
- [sFlow parameters and default values, on page 72](#)
- [sFlow Sampling, on page 73](#)
- [Configure sFlow, on page 73](#)
- [Revision History, on page 80](#)

sFlow Essential Concepts and Terms

This section helps you get familiar with the sFlow key terms and concepts:

- **Data source:** Location within a network device that can make traffic measurements. Examples are physical interfaces, VLANs.
- **Flow:** A Flow is defined as a set of IP packets passing a network device in the network during a certain time interval. All packets that belong to a particular Flow have a set of common properties derived from the data contained in the packet.
- **Flow record:** A Flow record is a set of key and non-key sFlow field values used to characterize flows. This record is created by inspecting packet headers and adding a description of packet information.
- **sFlow agent:** Entity inside the network device responsible for maintaining sFlow configuration, gathering the sampled flow and counters from one or more data sources in the router, packaging them in sFlow datagram format, and exporting them to the sFlow collector.
- **sFlow collector:** Application that receives the sFlow datagrams from one or more agents to perform further analysis and generate reports. The collector is external to the router.
- **Sampling rate:** Frequency that specifies how often packet sampling is performed, and determines how many packets (on average) that pass through the data source to generate a flow sample. A value of 100 means that on average, 1 out of 100 packets is randomly sampled to be exported.

- **Sampling interval:** Period at which counters will be polled for populating the counter sample in the sFlow datagram.
- **sFlow datagram:** User Datagram Protocol (UDP) datagram exported from sFlow agent to collector. The datagram contains information about the data source, one or more flow samples, and one or more counter samples.
- **Collector address:** IP and UDP port number. The default destination port number is 6343.

Flow Monitoring on Egress Interface

Table 14: Feature History Table

Feature Name	Release Introduced	Description
Flow monitoring on Egress Interface	Release 24.1.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100, K100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) You can now get precise insights into encapsulated and decapsulated data, prioritize critical outbound traffic, and ensure implementation of security measures. This is accomplished by activating egress flow monitoring on outbound interfaces within your network using sFlow. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH

Egress Interface Flow Monitoring enhances network visibility and control by prioritizing outbound traffic. This capability offers advanced monitoring and management of data exiting the network, providing a more comprehensive understanding of network dynamics. The key focus of this feature is to monitor packets that are either encapsulated or decapsulated through egress sFlow.

Encapsulated and decapsulated data monitoring in sFlow serves a crucial role in safeguarding sensitive information transmitted across the network. The process involves encapsulating data with an additional layer of information, enabling verification of its authenticity and integrity. This added layer makes it challenging for attackers to intercept or modify data during transmission. Conversely, decapsulation entails removing the encapsulated data layer, empowering network devices to analyze the information and take appropriate actions in real-time. This proactive approach aids in identifying and preventing attacks or anomalies, enhancing the overall security of the network.

How sFlow works

sFlow, a monitoring technology, operates by sampling data network traffic in real-time. However, it's important to note that sFlow doesn't encompass all network traffic, unlike Netflow.

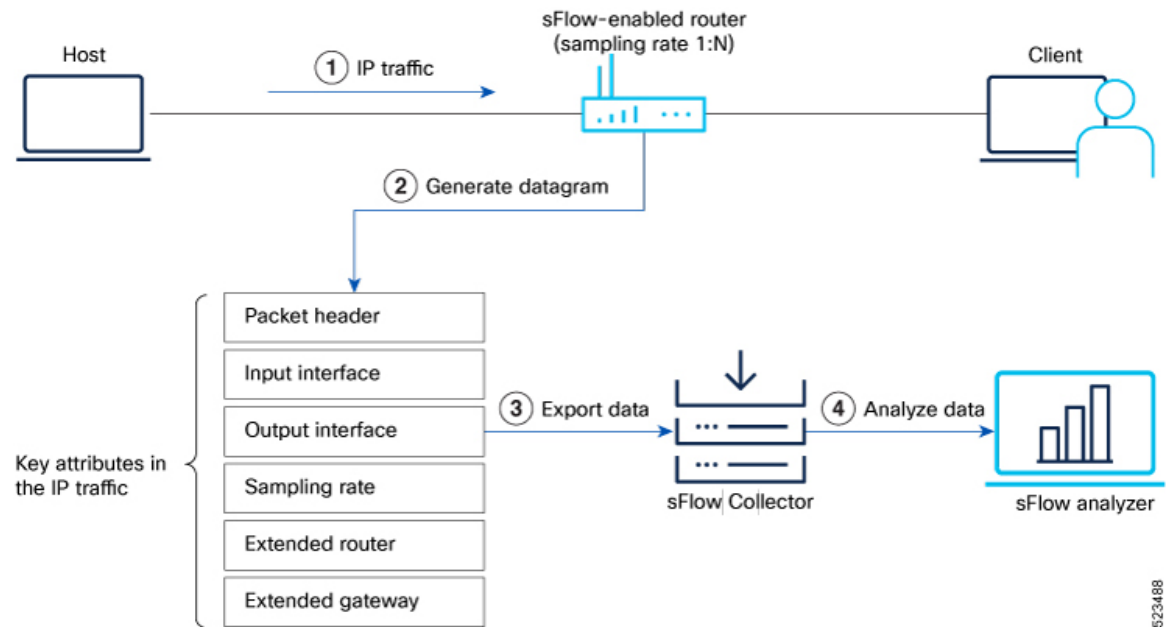
In the context of traffic monitoring, sFlow functions by disaggregating the flow pipeline. Devices within the network stream packet headers and metadata, which are subsequently transmitted as UDP datagrams to an external collector. This collector deciphers the packets and creates flow records. A notable feature of sFlow is its capability to export this data promptly, facilitating the creation of a near real-time representation of network traffic by the collector.

The advantage of this real-time traffic analysis is its ability monitor patterns and trends within the network, facilitate the automation of traffic engineering, and aid in making well-informed decisions when planning network capacity.

Recording of Packet Flows in sFlow

The packet in sFlow is recorded as follows:

Figure 5: Packet Flows in sFlow



1 Sampling

2 Datagram Generation

3 Data Export

4 Analysis and Reporting

In sFlow, the focus is on collecting sampled network traffic data rather than recording full packet flows. sFlow is designed to provide a statistical overview of network traffic by sampling packets and extracting relevant information for analysis.

Here's how sFlow handles the recording of packet flows:

1. **Sampling:** sFlow agent process in network devices sample packets based on a configured sampling rate. The sampling rate determines the percentage of packets that will be selected for analysis. For example, a sampling rate of 1-in-100 means that 1% of the packets will be sampled.
2. **Datagram Generation:** The sFlow agent generates datagrams that contain information about the sampled packets. These datagrams include details such as packet header, sampling rate, port numbers, protocol information, and various flow statistics.
3. **Data Export:** The sFlow datagrams are periodically exported from the sFlow agent to a designated sFlow collector or analyzer. The export can be done using protocols like UDP or TCP, and the datagrams are typically sent in a structured format like XDR.
4. **Analysis and Reporting:** Upon receiving the sFlow data, the sFlow collector or analyzer processes and analyzes the information. It aggregates the sampled data to provide statistical insights into network traffic, including top talkers, protocol distribution, traffic patterns, and other metrics.

sFlow Export with ECMP Load Balancing

Table 15: Feature History Table

Feature Name	Release Information	Feature Description
sFlow Export with ECMP Load Balancing	Release 25.1.1	Introduced in this release on: Fixed Systems (8700 [ASIC: P100]), 8010 [ASIC: A100])(select variants only*) This feature is now supported on: • 8712-MOD-M • 8011-4G24Y4H-I

Feature Name	Release Information	Feature Description
sFlow Export with ECMP Load Balancing	Release 24.4.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) You can now achieve sFlow packet load balancing across all ECMP paths to the collector. This feature utilizes pre-routing to gather nexthop interface and IP details for each packet, ensuring uniform distribution. By actively utilizing all paths, it provides more effective load balancing and improved network performance while maintaining path tracking. The feature introduces these changes: CLI: - The flow exporter-map command is modified to include the pre-route and all-ecmp-paths keywords. *This feature is supported on: 8212-48FH-M 8711-32FH-M 88-LC1-36EH 88-LC1-12TH24FH-E 88-LC1-52Y8H-EM

The sFlow ECMP Load Balancing feature enhances network routing by allowing routers to distribute sFlow packets across all available Equal-Cost Multi-Path (ECMP) routes to the sFlow collector.

The existing solution uses source-port entropy in the sFlow export UDP packets, introduced in IOS-XR software release 7.5.4, but it lacked the ability to track specific paths due to limited visibility of the outgoing interface. This new feature addresses these limitations by providing a pre-routing option, which gathers detailed nexthop information, including the output interface and nexthop IP address, for each packet. By actively utilizing all ECMP paths, the feature ensures efficient and uniform packet distribution, thereby improving routing performance, reliability, and network efficiency.

Benefits of ECMP Load Balancing

ECMP Load Balancing offers several key benefits:

- **Uniform Distribution:** Ensures even distribution of sFlow packets across all available paths, optimizing load balancing.
- **Enhanced Visibility:** Provides visibility into the nexthop details, overcoming the limitations of previous methods that lacked path tracking.
- **Improved Performance:** Actively uses all ECMP paths, enhancing network efficiency without compromising path tracking.

Mirror and stream drop packets using SPAN and sFlow

Table 16: Feature History Table

Feature Name	Release Information	Feature Description
Mirror and stream drop traps using SPAN and sFlow	Release 25.2.1	Introduced in this release on: Modular Systems (8800 [LC ASIC: P100]) The forwarding dropped trap feature enables system-generated notifications to be mirrored and forwarded to an sFlow collector when packets are dropped during the forwarding process. By leveraging SPAN and sFlow, this feature provides valuable insights into potential network issues and enhances network traffic monitoring and analysis by leveraging SPAN and sFlow.

Feature Name	Release Information	Feature Description
Mirror and stream drop packets using SPAN and sFlow	Release 25.1.1	Introduced in this release on: Modular Systems (8800 [LC ASIC: P100]) sFlow now supports buffer drop and forward-drop streaming, enhancing its capability to capture packets dropped by the Traffic Management (TM) buffer when full. This feature allows streaming of mirrored copies of these packets using SPAN, ensuring effective traffic monitoring even during process restarts or network failovers. Additionally, it mirrors forward-drop packets to capture and analyze packets dropped at router ingress, aiding in understanding blocked traffic types, identifying potential security threats, and optimizing network performance. The feature introduces these changes: CLI: The destination sflow command is introduced.

Mirror and stream drop packets using SPAN and sFlow

Starting from IOS-XR software release 25.1.1, the Mirror and Stream Dropped Packets using SPAN and sFlow feature enables the export of dropped packets using sFlow encapsulation. These packets are forwarded to an sFlow collector for in-depth analysis, alongside regular sFlow sampled packets. Differentiation is achieved using distinct UDP ports.

Key capabilities introduced by this feature include:

- Extends existing SPAN-based mirroring to support dropped packet streaming via sFlow.
- Uses distinct UDP ports to differentiate dropped and regular sFlow traffic.
- Extends SPAN functionality to support dropped packet streaming.
- Supports coexistence with regular sFlow traffic using separate exporter maps.

For more details about traffic mirroring feature, refer to the *Configure Traffic Mirroring* chapter in the *Interface and Hardware Component Configuration Guide for Cisco 8000 Series Routers*.

Mirror and stream drop traps using SPAN and sFlow

Starting from IOS-XR software release 25.2.1, the Mirror and Stream Drop Traps using SPAN and sFlow feature enables the export of forward-drop traps using sFlow encapsulation. A trap refers to a system-detected event where a packet is dropped due to specific conditions such as ACL denies, TTL expiry, or protocol violations. These dropped packets are encapsulated and streamed to an sFlow collector for real-time diagnostics and analysis.

- Provides real-time visibility into dropped traps.
- Uses sFlow encapsulation to export drop traps to collectors.

Types of drop packets

- **Buffer drop packets:** Buffer drop packets occur when the Traffic Management (TM) buffer reaches capacity and begins to drop incoming packets. This typically happens due to network congestion when the incoming traffic rate exceeds the buffer's processing capacity. The following scenarios lead to buffer drops:
 - **TM_EXACT_METER_DROP:** Occurs when the router drops packets because the incoming traffic rate exceeds the configured policer rate on the ingress interface.
 - **TM_STATISTICAL_METER_DROP:** Occurs when traffic is sent at a 100% line rate on an ingress interface, leading to packet drops.
- **Forward drop packets:** Forward drop packets are dropped during the forwarding process at the router ingress. This can occur due to several reasons, including:
 - **L3_IP_MC_PUNT_RPF_FAIL:** Occurs when an IP multicast RPF check fails.
 - **L3_ACL_FORCE_PUNT:** Occurs when layer 3 egress security measures or UDF ACLs are configured to punt packets to the host.
 - **L3_GLEAN_ADJ:** Packet drops during glean adjacency occurs when a router needs to forward packets but must first perform Address Resolution Protocol (ARP) resolution to obtain the next-hop MAC address.
 - **L3_DROP_ADJ:** Occurs when packets are dropped due to an inject up layer 3 lookup failure.
 - **L3_TTL_OR_HOP_LIMIT_IS_ONE:** Pertains to packets that have a Time-To-Live (TTL) or hop count of 1. These packets are typically dropped to prevent them from circulating indefinitely within the network, adhering to the principles of loop prevention.
 - **OAMP_BFD_MISMATCH_DISCR:** Occurs when there is a mismatch in the Bidirectional Forwarding Detection (BFD) session discriminator.

Starting from IOS-XR software release 25.2.1, the number of supported trap events has increased. Use the command **show controllers npu stats traps-all instance all local** to view them. The traps punted by the NPU are marked as (D*) and are supported.

Prerequisite for drop packet streaming

The following prerequisites must be met to utilize the stream dropped packets using sFlow.

- **Transition to sFlow:** Devices using NetFlow or IPFIX must transition to sFlow for regular sampling before utilizing the dropped packet feature, ensuring compatibility and consistency in data analysis.

- **Collector configuration:** A unified sFlow collector must be configured to handle both regular and dropped packet flows, utilizing different UDP ports to distinguish between the two streams.

Benefits of drop packet streaming

- Captures and forwards dropped packets to an sFlow collector, providing detailed insights.
- **Comprehensive Analysis:** Allows for simultaneous analysis of regular and dropped packet flows, offering a holistic view.
- **Troubleshooting:** Empowers network administrators to effectively identify and resolve issues, reducing downtime and enhancing performance.

Guidelines for Mirroring and Streaming Drop Packets

Use SPAN and sFlow to mirror dropped packets and traps for real time visibility and diagnostics to ensure that packet loss events are both observable and actionable

Configuration Best Practices

- Use a dedicated exporter map for drop traffic.
- Configure only one drop sFlow session per system to maintain export integrity.
- Use distinct UDP ports to differentiate drop traffic from regular sFlow exports.
- Do not modify DSCP marking for drop exports; it is fixed at 0.
- Ensure the export rate does not exceed 540 packets per second to avoid performance degradation.

System and Hardware Limitations

- **System Support:** Certain features like INJECT_UP_L3_LOOKUP_FAIL packet export are only supported on fixed systems, not modular ones.
- **ASIC Support:** TM_EXACT_METER_DROP is supported only on Q200 ASIC-based systems, excluding P100 ASIC-based systems.
- **Session Limitations:**
 - Only one global forward-drop session and one TM buffer drop session are supported for file (SPAN-to-file sessions), GRE (ERSPAN) tunnel interface, and sFlow destinations.
 - GRE tunnel interfaces and sFlow destinations cannot be configured together; they are mutually exclusive.
 - You cannot configure reachability for GRE and sFlow destinations through management interfaces.
- **ERSPAN Limitations:** Default encapsulation traffic class value is 0 for buffer and forward-drop packets, and ERSPAN counters are not updated for these packets.
- **sFlow export:** Drops resulting from inject packets (packets originating from the router) are not exported via the sFlow export facility.

These guidelines help maintain consistent monitoring, reduce configuration errors, and ensure dropped packets are captured and analyzed effectively.

sFlow parameters and default values

Table 17: Feature History Table

Feature Name	Release Information	Feature Description
Increased sFlow sample-header size	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I
Increased sFlow sample-header size	Release 24.4.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH
Increased sFlow sample-header size	Release 7.3.4	You can now increase the sFlow sampling size to 343 bytes of the incoming or outgoing packet header. This enhancement lets the router export a larger sample to the flow-analyzer tool, enabling the tool to provide more accurate network analytics. In earlier releases, you could configure up to 200 bytes.

The following table lists the sFlow parameters and default values that you can use when configuring sFlow on the router:

Table 18: sFlow Parameters and Default Values

Parameter	Value	Command
Sampling rate	Default value: 1 out of 10000 packets	sampler-map

Parameter	Value	Command
Sample header size	128 - 343 bytes (from Cisco IOS XR Release 7.3.4 onwards) 128 - 200 bytes (prior to Cisco IOS XR Release 7.3.4) Default value: 128 bytes	flow monitor-map
Counter poll interval	5-1800 seconds Default value: None	flow monitor-map
Collector port	Configurable. Default value: 6343	flow exporter-map

sFlow Sampling

The following methods are used in sFlow for capturing and analyzing network traffic:

- **Counter Sampling:** In the counter sampling method, only specific counters or statistics are sampled and collected for analysis. Instead of capturing and analyzing packets or flows, counter sampling focuses on monitoring and collecting information about specific network metrics or performance indicators. These metrics can include interface utilization, packet drops, CPU usage, memory usage, and other relevant statistics. Counter sampling provides a high-level view of network health and performance without the need to capture and analyze every single packet.
- **Flow Sampling:** Flow sampling, on the other hand, involves capturing and analyzing sampled network flows. A flow can be defined as a sequence of packets that share common attributes, such as source and destination IP addresses, port numbers, and protocol information. Flow sampling selects a subset of these flows for analysis. By capturing and analyzing flows, you can gain insights into traffic patterns, detect anomalies, and monitor performance. Flow sampling allows for more granular analysis of network traffic compared to counter sampling.

You can choose the method depending on the specific monitoring needs and objectives of the network.

Configure sFlow

This page explains how to configure sFlow for monitor network traffic using sampled data.

sFlow Guidelines and Limitations

General

- When egress sFlow is enabled, the Layer 2 information of ingress packets is captured instead of the egress interface. This behavior is not supported on the 88-LC1-12TH24FH-E and 88-LC1-52Y8H-EM line cards.
- sFlow samples are combined into UDP packets and forwarded to sFlow collectors for analysis. It's important to note that UDP, being a connectionless protocol, doesn't ensure the delivery of data.

Consequently, utilizing sFlow as a flow source could potentially lead to inaccurate representations of traffic volumes, bidirectional flows, and a reduction in alerting capabilities.

- Only one sampler configuration per router is allowed.
- A sampling interval of 1 out of 262,144 packets as the maximum is supported.
- Only one active sampler per router is supported. Multiple sampler maps that are not used in flow configuration are allowed.

Flow Exporter

- When sFlow is on a bundle with members located on different Line Cards (LCs), flows are exported with the same ifindex id for the bundle interface. However, they possess distinct sub-agent ids and sequence numbers.
- A maximum of 8 export destinations per monitor map for both IPv4 and IPv6 are allowed.

Flow Monitor

- Ingress sFlow is supported on Cisco 8200 and Cisco 8800 Series Routers.
- Egress sFlow is supported only on Cisco 8200 Series Routers.

Interfaces

- L3 interfaces, L3 bundle interfaces, L3 sub-interfaces, L3 bundle sub-interfaces, and L3 BVI interfaces are supported.
- Up to 2000 L3 interfaces are supported.
- Tunnel and Ethernet PseudoWire (PW) interfaces are not supported.
- ARP, multicast, broadcast, and IP-in-IP packets are excluded from the sampling process.

Loadbalance sFlow Traffic

- Cisco 8000 modular routers do not load balance sFlow traffic across multiple ECMP paths. When more than one path is available, the router selects only one. Typically, sending sFlow traffic over a single path is sufficient, and load balancing isn't necessary.

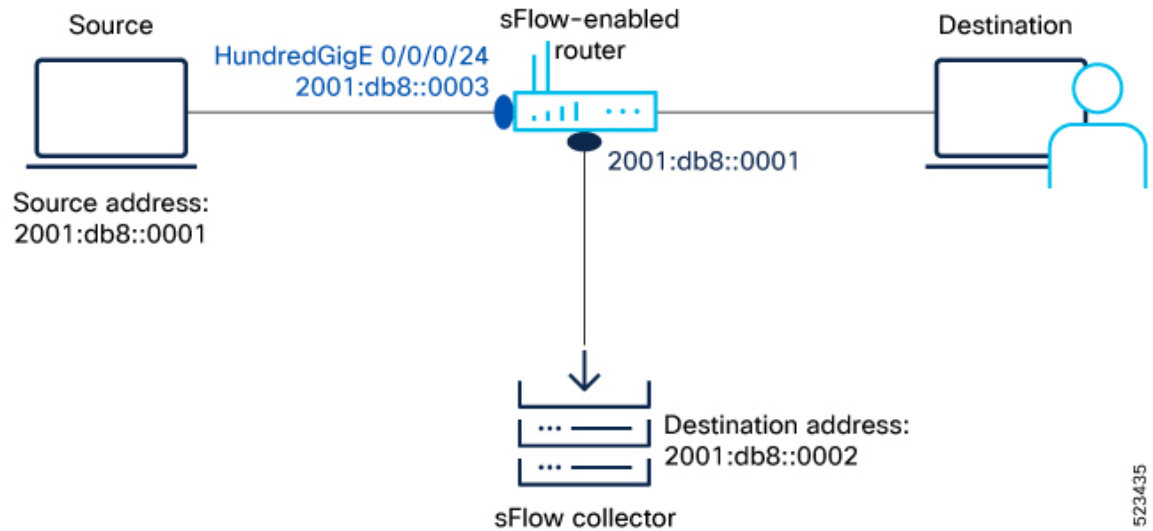
To enable load balancing across all ECMP paths, configure the router to use multiple UDP source ports for sFlow export packets. This method, known as "source-port entropy," uses a hashing-based algorithm to select paths, resulting in multiple ECMP paths being used. This approach does not provide statistics on the number of sFlow packets per path, as the sFlow process does not know the outgoing interface used. To configure multiple source ports, use the **transport udp source-port multiple** command.

Configuring sFlow

To enable this monitoring, it is necessary to configure the sFlow agent to use a sampling mechanism, forwarding traffic data from both ingress and egress ports to a centralized data collector, also referred to as the sFlow analyzer. The sampled data is forwarded using the version 5 export format. You'll find instructions for

configuring sFlow on the router in the following section. Let's use the following topology as a reference for configuring sFlow.

Figure 6: sFlow Configuration



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Procedure

- Step 1** First, let's gather the required details to enable sFlow on a router:
- The IP address of the source : 2001:db8::0001
 - The IP address of the sFlow Collector (Destination address): 2001:db8::0002
 - Interface of the router where we will enable sFlow: HundredGigE 0/0/0/24
 - sFlow version used to transport the data to the collector: version 5
- Step 2** Configure the Flow Exporter using the flow [flow exporter-map](#) command to specify where and how the packets should be exported.
- The following attributes can be configured while creating exporter map:
- Export destination IP address (IPv4 or IPv6 address). The same packets can be exported to multiple IPv4 or IPv6 destinations.
 - DSCP value
 - Source interface and its IP address
 - Transport protocol
 - UDP port number, where the collector listens to the packets
 - Maximum datagram length
 - Don't Fragment bit (DF-bit). The DF-bit within the IP header is supported only on IPv4 transport and determines whether the router is allowed to fragment a packet.

In this example, you create an exporter map called `EXP-MAP` to export the IPv4 packets to the destination address 192.127.10.1 using the UDP transport protocol:

```
Router(config)#flow exporter-map EXP-MAP
Router(config-fem)#version sflow v5
Router(config-fem)#packet-length 9000
Router(config-fem)#transport udp 6343
Router(config-fem)#source HundredGigE 0/0/0/1
Router(config-fem)#source-address 192.127.10.1
Router(config-fem)#destination 192.127.0.1
Router(config-fem)#dfbit set
```

• ECMP Load Balancing:

In this example, you configure ECMP load balancing option for sFlow.

```
Router(config)#flow exporter-map EXP-MAP
Router(config-fem)# pre-route all-ecmp-paths
Router(config-fem-pre-route)# all-ecmp-paths
```

• Stream Dropped Packets:

In this example, you configure forward-drop using SPAN on sFlow.

```
Router(config)#monitor-session mon1 ethernet
Router(config)#destination sflow exporter-map-name fem
Router(config)#drops
!
Router(config)#flow exporter-map fem
Router(config)#version sflow v5
!
```

Verify the Flow Exporter configuration using the `show flow exporter-map` command.

```
Router#show flow exporter-map EXP-MAP
Flow Exporter Map : EXP-MAP
-----
Id                : 1
Packet-Length     : 9000
DestinationIpAddr : 192.127.0.1
VRFName           : default
SourceIfName      : HundredGigE 0/0/0/3
SourceIpAddr      : 192.127.10.1
DSCP              : 0
TransportProtocol  : UDP
TransportDestPort  : 6343

Export Version     : sFlow v5
```

The Export Version: sFlow v5 indicates that the exporter map configuration is successful.

• Verifying ECMP Load Balancing:

```
Router#show flow exporter-map EXP-MAP pre-route path-index all location 0/0/CPU0
-----
Wed Dec  4 21:59:15.659 UTC

Exporter: EXP-MAP

Pre-route enabled: True

Enabled as default: False

Enabled in configuration: True
```

```

Platform capability: True

All ecmp paths: True

Statistics:

    Packets handled: 92

    Packets not handled: 1

Current Path list information:

    Reset time: 2024:12:04 21:53:07 UTC

    Path count:1

    Path index:0

    Path context:

        Nexthop Information:

            Output Interface Handle: 0x138

            Output Interface Name: FourHundredGigE0/0/0/1

            Nexthop-address: 30.0.0.2

        Statistics:

            Packets sent: 92

            Last Packet Sequence number: 92

```

• Verifying Dropped Packet Streaming:

```

Router(config)#show monitor-session status
Mon Dec 16 03:29:58.569 UTC
Monitor-session mon1
Destination sFlow - fem
=====
SW Mirrored Packet type  Dir
-----
All drops                Rx, Tx

Source Interface        Dir  Status
-----
Monitor-session default_monitor_session
Destination File - Packet collecting (always-on)
    Periodic write interval: 86400 seconds
    Maximum periodic capture capacity: 1GB
=====
SW Mirrored Packet type  Dir
-----
Packet-processing drops  Rx
Traffic-management drops Rx

Source Interface        Dir  Status
-----

```

- Step 3** Configure the Flow Sampler using `sampler-map sampler-map` command to define the sampling rate for flow samples, which determines how many packets (on average) that pass through the data source will generate a flow sample. In this example, you create a sampler map called `SAMP-MAP` to sample 1 out of every 4096 packets.

```
Router(config)#sampler-map SAMP-MAP
Router(config-sm)#random 1 out-of 4096
```

Verify the sampler map configuration using the `show sampler-map` command.

```
Router#show sampler-map SAMP-MAP
Sampler Map : SAMP-MAP
-----
Id:          1
Mode:        Random (1 out of 4096 Pkts)
```

In this example, the sampler map configuration is successful with a sample rate of 1 out of every 4096 packets.

- Step 4** Configure the Flow Monitor using `flow monitor-map` command to define the type of traffic to be monitored and the polling frequency. You can include one or more exporter maps in the monitor map. In this example, you create a monitor map called `MON-MAP` and include the exporter map `EXP-MAP` to record sFlow data at a polling interval of 120 seconds:

```
Router(config)#flow monitor-map MON-MAP
Router(config-fmm)#record sflow
Router(config-fmm)#sflow options
Router(config-fmm-sflow)#extended-router
Router(config-fmm-sflow)#extended-gateway
Router(config-fmm-sflow)#if-counters polling-interval 120
Router(config-fmm-sflow)#input ifindex physical
Router(config-fmm-sflow)#output ifindex physical
Router(config-fmm-sflow)#sample-header size 200
Router(config-fmm-sflow)#exporter EXP-MAP
```

You can export input and output interface handles if the ingress or egress interface is a bundle or a Bridge-Group Virtual Interface (BVI).

Verify the monitor map configuration using the `show flow monitor-map` command.

```
Router#show flow monitor-map MON-MAP
Flow Monitor Map : MON-MAP
-----
Id:          2
RecordMapName: sflow
ExportMapName: EXP-MAP
ExtendedRouter: Enabled
ExtendedGateway: Enabled
InterfaceCounters: Enabled
PollingInterval: 30 seconds
SampledHeaderSize: 200

Input ifhandle physical
Output ifhandle physical
```

In this example, the monitor map is configured successfully with the associated exporter map to monitor the interface counters at a polling interval of 30 seconds.

- Step 5** Apply sFlow on an interface using `flow datalinkframesection` command in Global Configuration mode.

In this example, you apply the monitor map `MON-MAP` and the sampler map `SAMP-MAP` on the HundredGigE 0/0/0/3 interface in the ingress direction to monitor the incoming packets:

```
Router(config)#interface HundredGigE 0/0/0/3
Router(config)#ipv4 address 192.127.0.56 255.255.255.0
Router(config)#ipv6 address FFF2:8:DE::56/64
Router(config)#flow datalinkframesection monitor-map MON-MAP sampler SAMP-MAP ingress
```

- Step 6** Enable sFlow on the RP (0/RP0/CPU0) or on the line card using [hw-module profile netflow sflow-enable](#) command.

```
Router(config)#hw-module profile netflow sflow-enable location 0/0/CPU0
```

- Step 7** Reload the line card using **hw-module reset auto** command.

```
Router#reload location 0/0/CPU0
```

With this configuration, sFlow is enabled on the line card.

- Step 8** Verify the statistics of exported traffic flow at the producer and exporter using [show flow platform producer statistics location](#) and [show flow exporter](#) commands.

Producer:

```
Router#show flow platform producer statistics location 0/0/CPU0
Netflow Platform Producer Counters:
IPv4 Ingress Packets:                0
IPv4 Egress Packets:                 0
IPv6 Ingress Packets:                0
IPv6 Egress Packets:                 0
MPLS Ingress Packets:               0
MPLS Egress Packets:                 0
IPFIX315 Ingress Packets:            0
IPFIX315 Egress Packets:             0
sFlow Ingress Samples:               100000
Drops (no space):                    0
Drops (other):                      0
Unknown Ingress Packets:             0
Unknown Egress Packets:             0
Worker waiting:                     0
```

Exporter:

```
Router#show flow exporter EXP-MAP location 0/0/CPU0
Wed June 21 04:21:36.263 UTC
Flow Exporter: EXP-MAP
Export Protocol: sFlow v5
Flow Exporter memory usage: 5247776
Used by flow monitors: MON-MAP

Status: Normal
Transport: UDP
Destination: 192.127.0.1 (6343) VRF default
Source: 192.127.10.1 (6331)
Flows exported: 50245 (9631004 bytes)
Flows dropped: 0 (0 bytes)

Packets exported: 7372 (19262008 bytes)
Packets dropped: 0 (0 bytes)

Total export over last interval of:
  1 hour: 7363 pkts
          9629960 bytes
          50236 flows
  1 minute: 12 pkts
            1392 bytes
            12 flows
  1 second: 0 pkts
```

0 bytes
0 flows

The sFlow configuration with flow and packet data is successful on the router.

What to do next

Analyze the traffic using the UDP datagram and sampled data collected at the sFlow collector.

This image shows an example of sampled data that has been collected at the sFlow collector.

1	0.00000000	110.0.0.1	110.0.0.2	sFlow	306 v5, agent 110.0.0.1, sub-agent ID 0, seq 25, 2 samples
2	5.00316998	110.0.0.1	110.0.0.2	sFlow	190 v5, agent 110.0.0.1, sub-agent ID 0, seq 26, 1 samples
3	5.04802656	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 27, 5 samples
4	6.00313678	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 28, 5 samples
5	9.93961608	110.0.0.1	110.0.0.2	sFlow	1430 v5, agent 110.0.0.1, sub-agent ID 0, seq 29, 6 samples
6	9.93962764	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 30, 5 samples
7	9.93963240	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 31, 5 samples
8	9.93963936	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 32, 5 samples
9	9.94382404	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 33, 5 samples
10	9.94400386	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 34, 5 samples
11	9.94401016	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 35, 5 samples
12	9.94446358	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 36, 5 samples
13	9.94781280	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 37, 5 samples
14	9.94822924	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 38, 5 samples
15	9.94831016	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 39, 5 samples
16	9.94857528	110.0.0.1	110.0.0.2	sFlow	1314 v5, agent 110.0.0.1, sub-agent ID 0, seq 40, 5 samples

Flow data length (byte): 136	
Header protocol: Ethernet (1)	
Frame Length: 118 bytes	
Payload removed: 4 bytes	
Header of sampled packet: 00fc8b539c0800d76eac930008004500006400230000fd01ff71640000025a000002080095d3391d0023abcdabcbcdabcbcd...	
Ethernet II, Src: 00:d7:6e:ac:93:00 (00:d7:6e:ac:93:00), Dst: 00:fc:8b:53:9c:08 (00:fc:8b:53:9c:08)	
Destination: 00:fc:8b:53:9c:08 (00:fc:8b:53:9c:08)	
Source: 00:d7:6e:ac:93:00 (00:d7:6e:ac:93:00)	
Type: IP (0x0800)	
Trailer: 000000000000	
Internet Protocol Version 4, Src: 100.0.0.2 (100.0.0.2), Dst: 90.0.0.2 (90.0.0.2)	
Version: 4	
Header length: 20 bytes	
Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))	
Type of service: 0x00 (None)	
Total Length: 100	
Identification: 0x0023 (35)	
Flags: 0x00	
Fragment offset: 0	
Time to live: 253	
Protocol: ICMP (1)	
Header checksum: 0xff71 [correct]	
Source: 100.0.0.2 (100.0.0.2)	
Destination: 90.0.0.2 (90.0.0.2)	
[Source GeoIP: unknown]	
[Destination GeoIP: unknown]	
Internet Control Message Protocol	
Extended router data	
Extended gateway data	

Revision History

Table 19: Feature History Table

Feature Name	Release Introduced	Description
System Alerts with sFlow	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I

Feature Name	Release Introduced	Description
System Alerts with sFlow	Release 24.4.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100, K100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) The syslog notifications with sFlow are now available on the following hardware. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH
System alerts related to sFlow	Release 7.5.3	The following syslog notifications are available with sFlow: • FLOW_SAMPLES_DROPPED - This alert is seen whenever the buffer becomes full with sampled flow data, either due to a high sampling rate or an increase in the traffic rate. • FLOW_SAMPLES_DROPPING_STOPPED - This alert is seen when the buffer reverts to its regular state. • BUFFER_SIZE_EXCEEDED - This alert signals that the flow monitor buffer has reached its capacity with sampled flow data, which could be a result of a low export rate limit or a high sampling rate. • BUFFER_EXCEEDING_STOPPED - This alert is seen when the flow monitor buffer reverts to its regular state.
Ingress sFlow enhancements	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I

Feature Name	Release Introduced	Description
Ingress sFlow enhancements	Release 24.4.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100, K100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) The Ingress sFlow enhancements added in Cisco IOS XR Release 7.3.3 now apply to the following hardware thus ensuring improved scalability and decreased volume of packets received. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH
Ingress sFlow enhancements	Release 7.3.3	The incoming sFlow packet offers the following enhancements to improve scalability and decrease the volume of packets received: • Expansion of sFlow datagram size—from 1500B to 9KB • Tunnel encapsulation—The packet header now supports an extended structure encompassing tunnel header information. The egress packet extracts the tunnel information during decapsulation. • sFlow collector indicates discarded packets and locally targeted packets at the output interfaces, in a specific format along with drop value.
sFlow on L2 interfaces	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I

Feature Name	Release Introduced	Description
sFlow on L2 interfaces	Release 24.4.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100, K100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) sFlow support on L2 interfaces is now extended on the following hardware to ensure efficient traffic monitoring on your network. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH
sFlow for L2 interfaces	Release 7.3.1	Ingress sFlow on an L2 interface is introduced. Support for sFlow existed in earlier releases.
Sampled flow	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I
Sampled flow	Release 24.4.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100, K100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) This feature is now supported on the following hardware thus allowing to monitor real-time traffic in your data networks and forward the sample data to the central data collector using the sampling mechanism in the sFlow agent software. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH

Feature Name	Release Introduced	Description
Flow monitoring on Egress Interface	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on: • 8011-4G24Y4H-I
Flow monitoring on Egress Interface	Release 24.1.1	Introduced in this release on: Fixed Systems (8200 [ASIC: P100], 8700 [ASIC: P100, K100])(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*) You can now get precise insights into encapsulated and decapsulated data, prioritize critical outbound traffic, and ensure implementation of security measures. This is accomplished by activating egress flow monitoring on outbound interfaces within your network using sFlow. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH
Sampled flow	Release 7.2.12	Sampled flow (sFlow) allows you to monitor real-time traffic in data networks. It uses sampling mechanism in the sFlow agent software to monitor traffic and to forward the sample data to the central data collector.

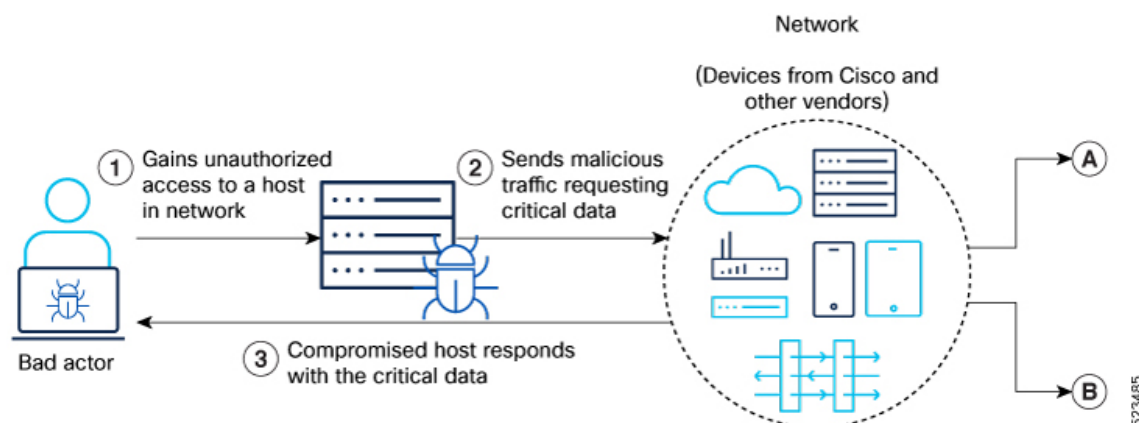


CHAPTER 7

Use Case: NetFlow and sFlow in Action

Here's a hypothetical use case illustrating a bad actor (attacker) sending malicious traffic and the network getting compromised:

Figure 7: Malicious activity in a network



1 Scenario A: Traffic Monitoring Without NetFlow and sFlow

2 Scenario B: Traffic Monitoring With NetFlow and sFlow

- Attack entry point—An Enterprise becomes the target of a cyber attack. The attacker employs various tactics to gain unauthorized initial access to the network.
- Generate malicious traffic— After the attacker identifies vulnerable devices as potential targets, they compromise a host and start generating malicious traffic and potentially launch DDoS attacks, to steal sensitive data, or take control of the network using these compromised machines as a platform.
- Breach data—The malicious traffic triggers a series of attacks within the network. With access to sensitive data, the attacker attempts retrieving critical data from the compromised network.

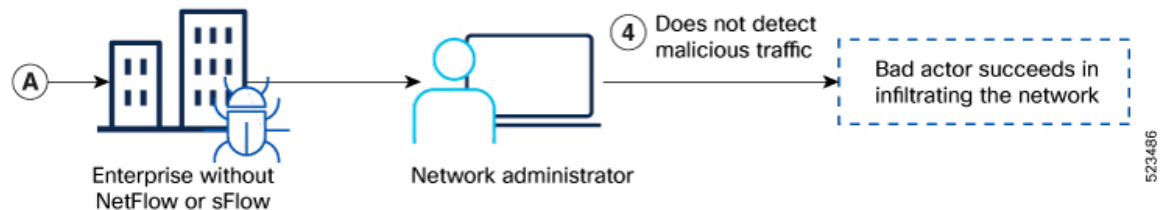
With this context, let's explore these two scenarios:

- [Scenario A: Traffic Monitoring Without NetFlow and sFlow, on page 86](#)
- [Scenario B: Traffic Monitoring With NetFlow and sFlow, on page 86](#)

Scenario A: Traffic Monitoring Without NetFlow and sFlow

In this particular situation, the enterprise had failed to implement any network traffic monitoring protocols such as NetFlow or sFlow.

Figure 8: Traffic monitoring without Flow data to identify malicious activity



Here is a high-level outline of the network's response to the attack:

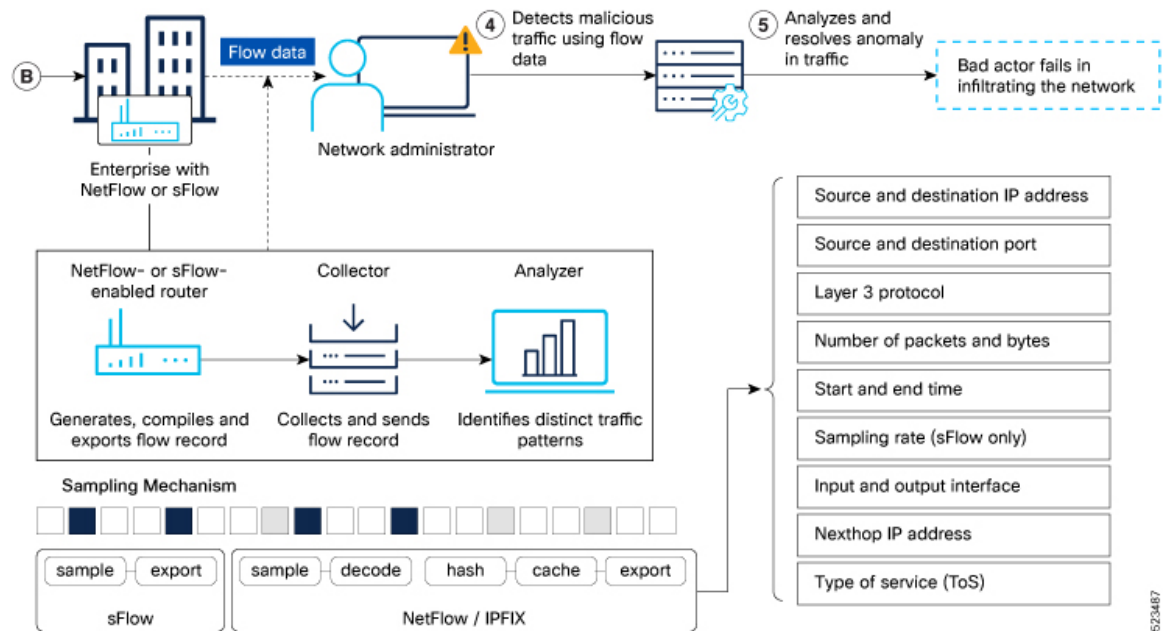
- Bypassed threat detection and response—The network administrator does not detect any unusual network patterns or intrusions immediately following the attack.
- Successful data breach—Consequently, the network is compromised through malicious traffic that gets undetected leading to loss of critical data and trust.

The overall network security posture is compromised due to lack of traffic monitoring mechanisms leading to poor visibility of the network and its functionalities.

Scenario B: Traffic Monitoring With NetFlow and sFlow

In this particular situation, the enterprise has implemented network traffic monitoring protocols such as NetFlow or sFlow.

Figure 9: Traffic monitoring workflow with Flow data to identify malicious activity



Here is a high-level outline of the network's response to the attack:

- **Flow data collection**—Routers enabled with NetFlow or sFlow capture and retain flow records of transmitted traffic. These records store essential metadata related to the traffic's journey, including source and destination domains, the count and volume of inbound and outbound packets, timestamps and so on. The recorded flow records are then sent to a designated collector.
- **Data analysis**—Utilize a NetFlow or sFlow analyzer or security monitoring tool to process and analyze the collected data. The tool can identify patterns and anomalies that may indicate a security threat, such as unusual traffic patterns, unexpected communication between hosts, or a high volume of traffic from suspicious sources.
- **Threat detection**—The analyzer applies algorithms and rules to detect potential threats based on the analyzed data. It can compare network traffic with predefined security policies. If a potential threat is detected, the analyzer generates an alert. This alert can be sent to the network administrator for further investigation.
- **Prompt investigation and responsive action**—Upon receiving the alert, the network administrator can investigate the identified threat. They can analyze additional logs, inspect packet captures, or perform other security measures to gather more information about the threat. Once the threat is confirmed, appropriate actions can be taken to mitigate the impact by blocking the malicious IP addresses and isolating affected hosts to prevent further harm.

By leveraging NetFlow and sFlow for threat identification, you can proactively detect and respond to security threats, enhancing the overall network security posture. It allows for early threat detection, and faster incident response, ultimately reducing the risk of a successful attack.



CHAPTER 8

YANG Data Models for NetFlow and sFlow

In this section, you'll learn to use the YANG data models to configure and retrieve the operational status of NetFlow and sFlow on Cisco 8000 Series Routers.

What You'll Find in This Section

Cisco IOS XR supports configuring NetFlow and sFlow using both traditional Command Line Interface (CLI) using commands as well as programmatically using YANG data models. In this section, you'll find references to supported YANG data models and an understanding about accessing and using these data models.

To get started with using these data models, see:

- [List of YANG Data Models for NetFlow and sFlow, on page 89](#)
- [Access Data Models, on page 90](#)
- [Get Started With IOS XR YANG Data Models, on page 92](#)

List of YANG Data Models for NetFlow and sFlow

Here is a list of YANG data models that you can use to configure and manage NetFlow and sFlow on the router:

Table 20: NetFlow and sFlow YANG Data Models

Cisco IOS XR Native Data Model	Unified Data Model	OpenConfig Data Model
Cisco-IOS-XR-traffmon-netflow-cfg	Cisco-IOS-XR-um-8000-hw-module-profile-netflow-cfg	openconfig-sampling-sflow.yang
Cisco-IOS-XR-ofa-netflow-oper	Cisco-IOS-XR-um-flow-cfg	



Note We recommend using Unified Data Models over Native Data Models

You can access the data models using one of these following options:

Access Data Models

You can access the data models using one of these following options:

Access Data Models From Router

To access data models directly from the router, you can follow these steps:

Procedure

Step 1 Enter the global configuration mode.

Example:

```
Router#configure
```

Step 2 Configure the NETCONF network management protocol to remotely configure and manage the router using YANG data models.

Example:

```
Router(config)#netconf-yang agent ssh
```

Step 3 Commit the configuration.

Example:

```
Router(config)#commit
```

Step 4 Establish a NETCONF session with the device and retrieve the capabilities information.

Example:

```
Router#show netconf-yang capabilities
Tue Sep 19 22:03:26.305 UTC
[Netconf capabilities]
```

```
D: Has deviations
```

Capability	Revision	D
urn:ietf:params:netconf:base:1.1	-	
urn:ietf:params:netconf:capability:candidate:1.0	-	
urn:ietf:params:netconf:capability:confirmed-commit:1.1	-	
urn:ietf:params:netconf:capability:interleave:1.0	-	
urn:ietf:params:netconf:capability:notification:1.0	-	
urn:ietf:params:netconf:capability:rollback-on-error:1.0	-	
urn:ietf:params:netconf:capability:validate:1.1	-	
http://cisco.com/ns/yang/Cisco-IOS-XR-8000-fib-platform-cfg	2019-04-05	
http://cisco.com/ns/yang/Cisco-IOS-XR-8000-lpts-oper	2022-05-05	
http://cisco.com/ns/yang/Cisco-IOS-XR-8000-platforms-npu-resources-oper	2020-10-07	
http://cisco.com/ns/yang/Cisco-IOS-XR-8000-qos-oper	2021-06-28	
http://cisco.com/ns/yang/Cisco-IOS-XR-Ethernet-SPAN-act	2021-03-22	
http://cisco.com/ns/yang/Cisco-IOS-XR-Ethernet-SPAN-cfg	2022-07-13	
http://cisco.com/ns/yang/Cisco-IOS-XR-Ethernet-SPAN-datatypes	2021-10-06	
http://cisco.com/ns/yang/Cisco-IOS-XR-Ethernet-SPAN-oper	2022-09-05	
http://cisco.com/ns/yang/Cisco-IOS-XR-aaa-aaacore-cfg	2019-04-05	
http://cisco.com/ns/yang/Cisco-IOS-XR-aaa-ldapd-cfg	2022-06-22	


```
http://cisco.com/ns/yang/Cisco-IOS-XR-aaa-ldapd-oper |2022-05-20|
http://cisco.com/ns/yang/Cisco-IOS-XR-aaa-lib-cfg |2020-10-22|
http://cisco.com/ns/yang/Cisco-IOS-XR-aaa-lib-datatypes
----- Truncated for brevity -----
```

By examining the capabilities, you can view the available data models for the software version installed on the router.

Access Data Models From Cisco Feature Navigator

To access data models from Cisco Feature Navigator, you can follow these steps:

Procedure

- Step 1** Go to [Cisco Feature Navigator](#).
- Step 2** If you have a Cisco.com account, click on the **Login** button and enter your credentials. If you don't have an account, you can click **Continue as Guest**.
- You will be directed to the Cisco Feature Navigator main page.
- Step 3** Click **YANG Data Models**.
- Step 4** Select the **Product** and **Cisco IOS XR Release** based on your requirement.
- The data models are listed based on type—Cisco XR native models, Unified models and OpenConfig models.
- You can use the search field to search for specific data model of interest.
- Step 5** Click the specific data model of interest to view more details.
- The data model is displayed in a hierarchical tree structure making it easier to navigate and understand the relationships between different YANG modules, containers, leaves and leaf lists. You can apply filters to further narrow down the data model definitions for the selected platform and release based on status such as deprecated, obsolete and unsupported nodes.
- You can also click the **Download** icon to export the data model information in Excel format.
- This visual tree form helps you get insights into the nodes that you can use to automate your network.
- The data models on Cisco Feature Navigator is regularly updated based on IOS XR release. If you encounter any problem or have suggestions for improvements, share your experience using [Send us your feedback](#) link.
-

Access Data Models From GitHub

To access the data models from GitHub repository, you can follow these steps:

Procedure

- Step 1** Go to the [GitHub](#) repository for data models.

On the repository page, you will find a list of folders based on IOS XR releases.

Step 2 Navigate to the release folder of interest to view the list of supported data models and their definitions. For example, if you want to access the data models for IOS XR release 7.10.1, click on the folder named `7.10.1`.

Inside the folder, you will find a list of YANG files representing different data models.

Step 3 Click on the YANG file you want to access to view its contents.

You can also click on the **Raw** button to see the raw code or use the **Download** button to download the file to your computer.

Each data model defines a complete and cohesive model, or augments an existing data model with additional XPaths. To view a comprehensive list of the data models supported in a release, navigate to the **Available-Content.md** file in the repository. The unsupported sensor paths are documented as deviations. For example, `openconfig-acl.yang` provides details about the supported sensor paths, whereas `cisco-xr-openconfig-acl-deviations.yang` shows the unsupported sensor paths for `openconfig-acl.yang` model.

Step 4 Repeat the above steps for other versions or data models of interest.

The GitHub repository for IOS XR data models is regularly updated based on release. You can also contribute to the repository by submitting pull requests, opening issues if you encounter any problems or have suggestions for improvements.

Get Started With IOS XR YANG Data Models

Here is a generic outline of the steps involved in programmatically configuring your router using YANG data models:

1. Enable network management protocol—Manage the router remotely using the protocols such as NETCONF or gRPC.
2. Install the necessary libraries and tools—Depending on the programming language you are using, you may need to install libraries or tools to programatically interact with the router. For example, if you are using Python, you might need to install the `ncclient` library.
3. Establish a session with the router—Use the programming language of your choice to establish a connection to the router using NETCONF or gRPC protocols. This involves providing connection parameters such as device IP address, username, password, and port number.
4. Retrieve the router capabilities—View the supported features and functionalities available on the router.
5. Create or modify configurations—Use YANG data models to create or modify the configuration on the router.
6. Apply the configuration—Push the updated configuration via the NETCONF or gRPC protocol to modify the router's running configuration to reflect the desired changes.
7. Validate the configuration—Verify that the changes are successfully applied. You can retrieve the running configuration or specific configuration parameters to ensure that the device is configured as intended.

For detailed instructions about using the data models, refer the *Programmability Configuration Guide for Cisco 8000 Series Routers*.



CHAPTER 9

Command-line Interface (CLIs) for NetFlow and sFlow

The Cisco Command Reference Guide serves as a comprehensive resource, offering a catalog of command-line interface (CLI) commands for configuring and verifying NetFlow and sFlow implementation.

What You'll Find in This Section

Cisco IOS XR supports configuring NetFlow and sFlow using both traditional Command Line Interface (CLI) using commands as well as programmatically using YANG data models.

In this section, you'll find:

- [Reference to Command Reference Guide, on page 93](#)

Reference to Command Reference Guide

The Cisco Command Reference Guide serves as a comprehensive resource, offering a catalog of command-line interface (CLI) commands for configuring and verifying NetFlow settings.

To view the list of supported commands, refer [Netflow Command Reference for Cisco 8000 Series Routers](#).

